Somebody that I used to know: Unexpected deaths and the heterogeneous value of a professional connection*

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Abstract

This paper studies the effect of an exogenous reduction in the size of a worker's professional network, due to the unexpected death of a past coworker, on labour market outcomes. Using administrative data from Sweden and a matched event study, I find that unexpectedly losing a connection lowers employment by 0.4 percentage points after three years, with slightly larger effects on earnings. Workers are less likely to work in the deceased connection's last workplace post-treatment and treated unemployed workers, in particular, take longer to find a job. I explore heterogeneity in the empirical value of a connection, using the Generalised Random Forest. Valuable connections were older and higher-earning before their death and had a smaller set of connections competing for information. Using these estimates to quantify the total value of a worker's network I find that removing networks would likely increase inequality. Employed workers at the bottom decile of earnings are 9.5 percentage points less likely to be employed after losing their network, compared to 1.8 percentage points for workers in the top decile.

Keywords: networks, job search, referrals, inequality

JEL codes: J30, J20, J62

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1 Introduction

The use of informal contacts is a central feature of job search. Networks help individuals overcome search frictions (Calvó-Armengol and Jackson, 2004, 2007; Beaman, 2012) and bilateral information frictions between workers and firms (Montgomery, 1991; Simon and Warner, 1992; Dustmann et al., 2016). Better networks therefore lead to higher earnings for the individual, through higher wages and reduced unemployment. Because search and information frictions are ubiquitous in job search, many have argued that unequal access to the high-quality networks that help overcome them is an important source of earnings inequality (Ioannides and Loury, 2004; Jackson et al., 2017; Jackson, 2025).[1]

However, the effect of job search networks on inequality also depends on the extent to which different workers can navigate the labour market without their networks. Survey evidence indicates that low-educated workers in fact rely more on their networks for generating job offers than higher-educated workers. In 2025, 43 per cent of workers with at most lower-secondary education surveyed in the EU cited "friends, relatives, or acquaintances" as the most effective method for finding a job, compared to only 19 per cent of those with at least some tertiary education, who were instead more likely to consider "job advertisements" the most effective method (Eurostat, 2026).[2] An individual's demand for job search information and the quality of their connections may therefore be negatively correlated across individuals. Which of these forces dominates will determine whether networks tend to lead to more or less equal labour market outcomes.

In this paper, I estimate the heterogeneous effect of a connection on earnings and employment and study how access to networks contributes to inequality in the labour market. I focus on a layer of workers' networks that is likely to be particularly important for job search: professional networks. Using full population matched employer-employee data for Sweden, I define an individual's relevant professional contacts as people with whom they shared a workplace in the preceding five years, but with whom they no longer work. To identify the effect of a connection,

[1] Networks are argued to cause inequality through two channels: *homophily* implies that high-wage workers are more likely to pass on information to high-wage contacts, with whom they share socio-demographic traits, than to low-wage contacts, while *segregation* across neighbourhoods, workplaces, or schools causes low-wage workers to have few high-wage connections to rely on.

[2] Such educational gradients are a recurring finding in different contexts in time and space, see e.g. Corcoran et al. (1980); Pellizzari (2010) for similar survey evidence on job-finding methods.

I rely on the natural experiment created by the unexpected death of former coworkers. I use information on the cause of death to identify 4,129 unexpected deaths of past coworkers during the period 1991–2010, which affect 97,552 treated individuals.

In the first part of the paper, I apply a matched event study approach to estimate the average effect of a death on subsequent employment and earnings. I compare treated individuals, affected by the true death of a past coworker, with control individuals, affected by the placebo death of a past coworker, matched on an estimated propensity score. If treated and matched control individuals' earnings and employment follow parallel trends in the absence of treatment, this design identifies the average treatment effect on the treated of a past coworker death. I draw on the classic model of vacancy sharing on a network of Calvó-Armengol and Jackson (2004) to show that this causal effect can be interpreted as the total effect on an individual of losing a connection, given both the direct effect of losing a connection on the treated individual and any spillover effects on the individual through mutual connections of the treated and deceased individual.[3]

Losing a past coworker has a negative average effect on workers' labour market outcomes. Three years after the death, annual earnings fall by around 0.6 per cent (p-value = 0.066) relative to earnings in the control group, or 1,100 SEK (approximately 105 EUR in 2018 prices). This is largely due to a reduction in the employment probability, which falls by around 0.4 percentage points (p-value = 0.017).

I establish that the network channel is likely to be the main mechanism through which the death of a past coworker affects earnings and employment. First, initially unemployed workers are particularly negatively affected by the loss of a past coworker. They are nearly 2 percentage points less likely (p-value = 0.002) to be employed three years after treatment, due to a lower job-finding rate. Second, the treatment leads to a 0.1 percentage point reduction (p-value $<$ 0.001) after three years in the probability of working in the deceased individual's last workplace, conditional on the deceased individual being employed. In contrast, there is a small, imprecisely estimated positive effect on the probability of working at the workplace the treated

[3]The conceptual framework I present only includes search frictions. More generally, the effect identified by the matched event study is also a total effect in the sense that it sums up the effect of the treatment on both the search and information frictions the treated worker faces and any endogenous changes in, e.g., search effort in response to the treatment.

individual last shared with the deceased individual, where the connection was formed. Taken together, these findings suggest that treated individuals face greater search frictions after the loss of a connection.

I consider various alternative mechanisms that might explain the estimated average effect. The death of a *current* coworker increases the firm's demand for surviving employees' labour, raising their earnings (Jäger and Heining, 2022). I find that the death of a *former* coworker has negative employment and earnings effects, implying that any increase in the deceased worker's employer's labour demand does not affect the deceased individual's professional network. I also find that the negative earnings effects emerge gradually, whereas grief following the death of a close family member has been found to have immediate negative effects on employment and earnings (van den Berg et al., 2017; Jensen and Zhang, 2026). Finally, I show in the second part of the paper that the largest negative effects are observed following the death of older, higher-earning individuals, who are more likely to have better information to pass on. This pattern suggests that the effect is unlikely to be due to changes in either the wage–amenity or labour–leisure tradeoff workers make as they update their perceptions of their own mortality risk.

In the second part of the paper, I turn to assessing the contribution of networks to unequal labour market outcomes. I use a machine learning algorithm, the Generalised Random Forest (GRF, Athey et al., 2019) to empirically characterise heterogeneity in the effect of losing a connection. The GRF allows me to simultaneously consider the large number of possible sources of heterogeneity in the value of a connection that have been proposed in the literature without overfitting the data. First, I find that negative treatment effects are concentrated in the bottom quintile of dyads, ranked on treatment effects. In the remaining dyads there is a null effect on the treated individual from the death of their past coworker. Second, both dyad members' characteristics matter for the size of the treatment effect. Treatment effects are negative when the treated individual is lower-earning and less strongly attached to the labour market and when the deceased past coworker is older, higher-earning, and more strongly attached. Network structure, as well as other demographic characteristics including gender or education, or homophily along these dimensions, are weaker predictors of the effect of losing a connection.

Finally, I relate the estimated patterns of heterogeneity to inequality in employment out-

comes. I use the output of the GRF to predict the effect of losing a connection for each individual's entire set of connections. Heterogeneity in the average across a worker's connections of the marginal value of a connection follows a similar pattern to the heterogeneity observed in the individual connections use to train the GRF. I also estimate the total value of workers' networks, allowing the value of remaining connections to vary as individuals are removed from the network. Completely removing an unemployed worker's network would reduce their re-employment rate by 21 percentage points after three years. Employed workers in the bottom earnings decile would be 9.5 percentage points less likely to be employed after three years, while workers in the top earnings decile would only be 1.8 percentage points less likely to be employed. Potential differences in access to high-value connections do not, therefore, undo my earlier findings on the relationship between baseline economic outcomes and heterogeneity in the value of a single connection. Workers at greater risk of unemployment are more harmed by the loss of their networks. I conclude that the use of professional networks in job search tends to reduce inequality.

The paper contributes first of all to the empirical literature on the average effect of professional networks on labour market outcomes. Existing studies have isolated the effect of networks on different frictions for specific populations. Higher-quality professional networks reduce search frictions for both employed workers (Caldwell and Harmon, 2019) and unemployed workers (Cingano and Rosolia, 2012; Glitz, 2017), while workers hired through referrals from former coworkers tend to do better, as the hiring firm faces fewer information frictions (Hensvik and Skans, 2016; Glitz and Vejlin, 2021). My approach instead estimates the net effect of losing a connection, combining its effects on both search and information frictions. The fact that the negative earnings effect can be mostly attributed to reduced job finding of unemployed workers suggests that the quantitatively more important effect of professional networks in my setting is to reduce search frictions for the unemployed.[4]

The second contribution of the paper is to the literature on heterogeneity in the value of social connections. Previous empirical work has considered the importance of individual

[4]My estimated effects also incorporate workers' endogenous responses to losing a connection, due to, e.g., more costly search effort (Caria et al., 2023). Income effects from more costly search are likely to be larger for workers drawing wages from a better wage distribution. This would lead to greater attenuation of the substitution effect arising from more costly search, and smaller employment effects for high-earnings individuals, which is also consistent with my findings.

characteristics in isolation, including race (Pedulla and Pager, 2019), nativity (Åslund et al., 2024), gender (Beaman et al., 2018), education levels (Hensvik and Skans, 2016), or age (Loury, 2006). Focusing on these socio-demographic characteristics is motivated by a theoretical claim that homophily, or similarity between individuals, is an important predictor of the value of a connection (Bayer et al., 2008; Currarini et al., 2009; McPherson et al., 2001). I consider a wider set of potential predictors of the value of a connection and use a method, the GRF, that allows me to assess the relative importance of different characteristics. I find that economic variables, such as past earnings and employment, are more important predictors of the value of a connection than socio-demographic variables such as gender, education, or nativity. Connections are particularly valuable for workers less attached to the labour market and valuable connections are those that are themselves well-placed in the labour market.

My findings also highlight the modest importance of network structure in determining the value of a connection. Theoretical papers have argued that connections with whom workers share few common ties, known as weak ties, are particularly valuable, since these individuals are more likely to provide new information (Granovetter, 1973), or because there is less competition to receive information from these connections (Calvó-Armengol, 2004; Calvó-Armengol and Zenou, 2005). In contrast, I find that tie strength is less important than economic characteristics and that some degree of overlap between two individuals' connections may in fact be associated with a higher probability of information flowing between those individuals, consistent with more recent empirical evidence on weak ties (Gee et al., 2017a,b; Rajkumar et al., 2022). Nevertheless, I do find that valuable connections have fewer connections of their own to pass on information to, consistent with job vacancy information being rival.

Finally, I contribute to our understanding of the contribution of professional networks to labour market inequality. Theory suggests that segregation in networks will lead to persistent differences in earnings and employment between groups (Calvó-Armengol and Jackson, 2004, 2007), particularly in the presence of homophily in information sharing (Okafor, 2025; Bolte et al., 2026). However, empirical work explicitly relating job search on networks to inequality has focused on specific segments of the labour market such as unemployed workers (Eliason et al., 2023) or high-skilled workers (Cullen and Perez-Truglia, 2023; Zimmerman, 2019), or on specific dimensions of inequality, in particular segregation and inequality on race or nativity (Pedulla

and Pager, 2019; Miller and Schmutte, 2021; Åslund et al., 2024). My approach instead speaks directly to the more general relationship between the use of networks in the labour market and inequality in earnings or employment.

I find that the total employment effect of removing a worker's network is largest for the unemployed and lower-earning individuals and that the use of networks in job search may therefore reduce inequality in labour market outcomes. In this respect, the paper is particularly closely related to two recent contributions to the literature. Lester et al. (2026) use US survey data and a structural model to show that low-ability workers rely more on their network for generating job offers than high-ability workers. I use a natural experiment and full-population administrative data and similarly find that low-earning workers are more dependent on their network for generating job offers. The similarity holds despite my focus on professional networks, which Lester et al. find are less valuable than family and friends for generating offers for low-ability workers. San (2026), and before that Kramarz and Skans (2014), also use a natural experiment and full population data for respectively Israel and Sweden to document socioeconomic gradients in the effects of parental networks on school-leavers' job finding. I use a different identification strategy and find a similar socioeconomic gradient in the effect of professional networks across the working-age population more broadly.

The paper proceeds as follows. I outline a conceptual framework of information sharing in a network that guides the empirical analysis in Section 2. In Section 3 I describe the data and sample used. In Section 4 I report event study estimates of the average effect of a lost connection on earnings and employment and explore mechanisms. In Section 5 I report and interpret GRF estimates of heterogeneity in the value of a connection and apply these to the discussion of how networks contribute to inequality in Section 6. Section 7 concludes.

2 Conceptual framework

2.1 Network structure and heterogeneity

In this section, I outline a model of employment with job information sharing on a network, drawing on the seminal model of Calvó-Armengol and Jackson (2004). This conceptual frame-

work is not used for structural estimation. Instead, its purpose is to guide the empirical analysis, clarifying what theoretical object is identified by the loss of a former coworker and illustrating why the effect of losing a past coworker is necessarily heterogeneous.

The simplest version of the model contains n identical agents, connected by a graph, where $g_{ij} = 1$ if agents i and j are connected, $g_{ij} = g_{ji}$, and $g_{ii} = 1$. Time is discrete, indexed by t; at the end of each period, agents are either employed, $s_{it} = 1$, or unemployed, $s_{it} = 0$. The vector $\mathbf{s}_t$ describes the employment status of all agents at the end of period t. Each period unfolds as follows. Agents enter period t with employment status $\mathbf{s}_{t-1}$. Each individual receives a job offer with probability $0 < a < 1$. If an agent is unemployed, they take the job. If an agent is employed, they pass the job on to a uniformly randomly chosen unemployed connection, if they have one. If agents receive multiple offers, they randomly choose one; unused offers are discarded. At the end of the period, all employed agents face an exogenous job destruction probability $0 < b < 1$.[5]

In Section 4, I use an event study to estimate the employment effect of losing a connection k. I model the treatment as equivalent to setting $g_{kj} = 0$ for all j. Denote employment probabilities under the original network structure as $P(\cdot)$ and when k has been removed from the network as $P_{-k}(\cdot)$. I normalise the time index so that the agent is removed from the network between the end of $t = 0$, the last pre-treatment period, and $t = 1$. I define the l period-ahead causal effect on i of removing k from the network as

$$\tau_{ik}^{l} \equiv P_{-k}(s_{il} = 1) - P(s_{il} = 1).$$

In Online appendix A, I show that the l period-ahead causal effect has two components:

$$\tau_{ik}^{l} = \sum_{\mathbf{s}_{l-1}} \Bigg\{ \underbrace{\big[P_{-k}(s_{il} = 1|\mathbf{s}_{l-1}) - P(s_{il} = 1|\mathbf{s}_{l-1})\big]P(\mathbf{s}_{l-1})}_{\textit{direct effect}} + \underbrace{\big[P_{-k}(\mathbf{s}_{l-1}) - P(\mathbf{s}_{l-1})\big]P_{-k}(s_{il} = 1|\mathbf{s}_{l-1})}_{\textit{indirect effect}} \Bigg\} \quad (1)$$

[5]The model abstracts from uncertainty about match quality and the effect of referrals on productivity, wages, and turnover (Montgomery, 1991; Simon and Warner, 1992; Galenianos, 2013; Dustmann et al., 2016). This is unlikely to be a limitation, since the mechanisms driving heterogeneity in vacancy provision to connections will largely be the same as the mechanisms driving heterogeneity in information provision to employers.

The first term, the direct effect, measures the average change in information agent i receives from their network, in a given period and across different states, when agent k is removed from the network.[6] In states where i is unemployed and k is employed in $l-1$, i experiences a negative *information effect*. Removing k from the network makes i less likely to receive information about a vacancy, decreasing the probability of employment. In states where both i and k are unemployed, i experiences a positive *competition effect*. Removing k from the network reduces the competition i faces from k for vacancies passed on by their mutual connections. The direct effect is a weighted average of the information and competition effects, where the weights $P(\mathbf{s}_{l-1})$ measure how frequently one or other effect would arise if k stayed in the network through $l-1$ and were only removed before the start of the period l.

The second term of Equation (1), the indirect effect, measures the change in the distribution across states when k is removed from the network. This effect is indirect, since it measures how all members of k's connected set are exposed to changes in each other's employment status through period $l-1$ when k is removed from the network. For this reason, I also refer to the indirect effect as an *insurance effect*, as it captures changes over time in the network's ability to insure its members against employment shocks. Its sign is ambiguous; if removing k shifts the network towards states $\mathbf{s}_{l-1}$ where i has a lower chance of being employed at the end of l, the insurance effect will be negative.[7]

In Section A of the Online appendix, I study how the l period-ahead employment effect varies with the topology of the network, the only source of heterogeneity between agents in this simple model. I show that the information effect is weakly decreasing in absolute value in the number of i's connections and strictly decreasing in the number of k's unemployed connections. The competition effect has an ambiguous relationship with the number of either i or k's connections and this relationship depends on the degree of overlap between i and k's set of connections. The competition effect is decreasing in the number of i's employed connections who are not also connections of k and has an ambiguous relationship with the number employed connections of

[6] Agent k's employment status in $l-1$ is a counterfactual; it is the employment status k would have had in $l-1$, had they remained in the network.

[7] The direct effect therefore captures the contemporaneous average employment effect of losing a connection, keeping the employment history constant. The indirect effect captures how the direct effect propagates through k's connected set over time, altering the employment history of agents in that set through period $l-1$.

both i and k. Finally, both the weights $P(\mathbf{s}_{l-1})$ and the indirect effect are the solution to a combinatorial problem that depends on the entire network structure and do not admit a general representation, other than in the special case where $l = 1$.

I therefore generate a network and simulate the effect of removing different individuals from the network, given different initial employment states, setting $a = 0.3$ and $b = 0.05$. The network is shown in Figure C.13, the details of the simulation are in Section A.3 of the Online Appendix. The results of the simulations suggest, for the parameter values used, that: (i) information effects dominate in the medium-term ($l \leq 6$) and the average causal effect is negative; and (ii) the l period-ahead causal effect is decreasing in absolute value in the number of i's connections, consistent with the comparative static results for the information effect, and does not have a monotonic relationship with either the number of k's connections, or the degree of overlap between i and k's connection set, but depends on both, as in the comparative static results for the competition effect.

2.2 Heterogeneity in other characteristics

Until now, I have deliberately limited heterogeneity in the simple framework to heterogeneity arising from network structure and initial employment status. However, the literature has highlighted many other potential sources of heterogeneity that may influence the causal effect of removing a connection, τ_{ik}^{l}.

First, the literature on job loss (Sullivan and Von Wachter, 2009; Seim, 2019; Athey et al., 2026) and the literature on aggregate employment dynamics (Hall and Kudlyak, 2019; Doniger, 2026; Cairó and Cajner, 2018) highlight that job finding rates a and separation rates b vary across worker types. In the model of information sharing presented here, workers with a higher individual job finding rate, now denoted a_i, will experience smaller information and competition effects, in absolute value, from losing a connection. Workers with a higher separation risk, denoted b_i, will experience smaller information and competition effects, but will more often find themselves in states where $s_{il-1} = 0$, where these effects receive a non-zero weight. Losing a connection with a higher job finding rate, now denoted a_k, will in general lead to a larger information effect and, if insurance effects dominate competition effects in k's connected set, a higher a_k is likely to have a negative relationship with the insurance effect. Losing a connection

with a lower separation rate, b_k, will shift the distribution of states, $P(\mathbf{s}_{l-1})$, placing more weight on states where k's connections experience an information effect, and is likely to have a negative relationship with the insurance effect.

Second, jobs are unlikely to be homogeneous. Generalised models of vacancy sharing in networks assume that offers come with a wage, workers keep vacancies with a higher wage than their current wage, and pass on vacancies to workers with a lower current wage than the offered wage (e.g. Calvó-Armengol and Jackson, 2007; Beaman, 2012). The state space in these models is therefore a vector of wages rather than employment statuses. Furthermore, workers may receive wage offers from different distributions, reflecting differences in human capital. The predictions of models with wages will be less sharp than the simple model presented here, however, workers drawing wages from higher wage distributions are in general less likely to experience either competition or information effects, while losing a connection who draws wages from a higher offer distribution will increase the magnitude of the information effect and reduce the magnitude of the competition effect.

Finally, workers are unlikely to uniformly randomly share information about vacancies. A large literature on information sharing in networks highlights that homophily, the degree of similarity between two individuals, is an important determinant of the propensity to share information (McPherson et al., 2001; Currarini et al., 2009, 2010; Bolte et al., 2026). In the simple framework used here, homophily in information sharing makes the probability that an employed worker passes on a vacancy to an unemployed contact a function of the socio-demographic characteristics of both individuals. When agents preferentially pass vacancies to similar individuals, it is likely that both the information and competition effect of removing a connection are increasing in absolute value in the degree of similarity between individuals i and k. The information effect is likely to decrease in absolute value with the average similarity between i and their remaining connections, while the competition effect is likely to decrease in the average similarity between i and the connections they do not share with k.

The framework presented here has two main implications. First, removing an individual k from the network has both a direct effect on the information flowing to i from their connections in each state of the world and an indirect effect on which states of the world are more common. The event study presented in Section 4 identifies the net effect of removing agent k from the

network, rather than the underlying parameters describing the job market, e.g. a and b, or the competition and information effects per se. Second, the employment effect of removing a connection is naturally heterogeneous, potentially along many dimensions at the same time, and different sources of heterogeneity are likely to interact with each other. Rather than try to study each potential mechanism in isolation, I will use a method that allows me to consider observable proxies for many sources of heterogeneity simultaneously in Section 5.

3 Data and sample construction

3.1 Data sources

The empirical analysis is based on administrative data for the entire Swedish population 16–64 years old during the period 1985–2018. I draw on the following administrative registers, held by Statistics Sweden.

I collect information on employment and earnings from the employment register (*Register-baserad arbetsmarknadsstatistik*, RAMS) for 1985–2018, which contains a link between gainfully employed persons and the firms and workplaces at which they work. Through statutory income statements, filed by employers to the Swedish Tax Agency, both the employee and the firm and establishment is identified. The employment register also includes information on job spells which I use to identify past coworkers, discussed in greater detail below.

I collect information on gender, birth year and level of education for the entire population from the SYS database for 1985–1989 and LOUISE for 1990–2018; both are integrated databases on education, income, and employment previously compiled by Statistics Sweden from various administrative registers for research purposes. I collect the county of birth for the Sweden-born population and the country of birth or a region of birth for the foreign-born population from a separate table based on the Total population register (*Registret över totalbefolkningen*, RTB). I collect information on the cause of death from the Cause of death register (*Dödsorsaksregistret*), available 1985–2010.

The data requirements for constructing the network variables, knowing the cause of death, and observing individuals for a sufficient window on either side of the death of a coworker mean

that I analyse the effect of deaths occurring in the years 1991–2010.

3.2 Defining professional connections

To identify an individual's set of relevant job search contacts, I proceed in several steps. First, employment spells, which are reported in months, are collapsed to a single main job for each individual and year. This is defined as the workplace where the individual is employed in the month of December and from which the individual has the highest annual earnings. If an individual is not employed anywhere in the month of December, no main job is defined for that year. All subsequent analyses build on the annual panel of main jobs to construct firm characteristics, individual networks, etc. The only exception is individual annual earnings, which I calculate using all employment spells in a calendar year, deflate to 2018 Swedish crowns (SEK), and winsorise at the 99th percentile.

To identify an individual's professional connections in year $t-1$, the year before they are potentially affected by the death of a connection, I first identify their workplaces during the past five years, i.e. in $[t-6, t-2]$. I then define an individual's connections as all coworkers from the past five year years who are neither (i) working with the focal individual in year $t-1$, nor (ii) working in the common past workplace. Connections may therefore be unemployed or working in a new workplace. Given the definition of a connection I use, the size of an individual's professional network will be a function of the following factors:

1. Attachment to the labour market, measured as an individual's employment rate during the network-building phase;
2. Own mobility, which affects the number of distinct workplaces during the network-building phase;
3. Average size of the workplace during the network-building phase.
4. Past coworkers' mobility, i.e. their propensity to leave a common past workplace

An increase in any of these factors will on average increase the size of an individual's network.

Firms operating on multiple sites, e.g. firms in construction or cleaning services, and municipal firms do not have defined workplaces. Certain workplaces may also be very large. I impose

the following restrictions on individuals in my sample: (i) their largest past workplace employed at most 1000 individuals; and (ii) their employers during the network building phase always have well-defined workplaces. This is both for computational reasons, so that the network of past coworkers in the full population does not contain billions of links, and conceptual reasons, to ensure that the focal individual likely had some interaction with their past colleagues. Prior evidence in the Swedish context has shown that the probability that a past coworker leads to a hire in their current firm is decreasing in past workplace size and falls to zero when the past workplace employed more than 100 workers (Eliason et al., 2023).

3.3 Treated dyads

Next, I identify unexpected coworker deaths. I will refer to four types of individuals: treated and control individuals, collectively called focal individuals, and deceased and placebo individuals, collectively called past coworkers. Here and in the matching procedure described below, an observation is a *directed dyad*, i.e. a directed edge in the graph of former coworkers, made up of either a treated individual and their deceased past coworker or a potential control individual and placebo deceased past coworker (who does not die in the treatment year). The treatment year t is the year in which a past coworker dies, the set of connections is constructed in year $t-1$ using the procedure defined in Section 3.2. I impose the following restrictions on the included treated and control individuals:

1. they are aged 19–54 in year $t-1$ (I exclude older workers so that early retirement is not a possible response to the death of a past coworker);
2. they do not die themselves in year t;
3. they are affected by the death of at most one past coworker in year t;
   (a) each treated individual is affected by the death of exactly one past coworker, the deceased individual.
   (b) each control individual is not affected by the death of any past coworkers in t.

I do not place any restrictions on treated and control individuals' employment status in year $t-1$. Deceased or placebo individuals satisfy the following restrictions:

1. they are aged 19–59 in year $t-1$;

2. their own set of connections is observed in $t-1$ (i.e. they have not recently worked in a firm without well-defined workplaces)

3. they have moved on from the last workplace they shared with the treated/control worker by year $t-1$, either by entering unemployment or by moving to a new, well-defined workplace.

The final pre-matching sample is made up of all directed dyads where the focal individual satisfies the first set of restrictions and the past coworker satisfies the second set of restrictions. Finally, given prior evidence that the value of a professional connection decays rapidly in prior workplace size, I keep only dyads where the connection was formed in a workplace employing at most 100 workers.[8]

I plot annual earnings for individuals who die of different causes in year t and individuals who do not die in t in Figure C.1, pooling observation years t in 1991–2010. Deaths from chronic illness are preceded by declining earnings, which may already have spillover effects on treated individuals' earnings or job search behaviour before the year of death. To prevent such anticipatory effects, I use information on the cause of death to restrict the sample of deceased individuals to those whose death was unexpected, exclude deaths due to intentional self-harm or assault.[9]

Given these restrictions, the final unmatched sample includes 97,552 treated dyads, made up of 97,552 treated individuals connected to 4,129 deceased past coworkers.[10] I report the distribution of causes of death over events in Figure 1a; just under half of deaths are due to transport accidents, essentially road accidents. I plot the frequency of deaths over years in Figure 1b. There is a noticeable spike of excess deaths in 2004 and a smaller spike in 1994.

[8]Note that this restriction only applies to dyads included in my sample. When defining an individual's network, I continue to include all workplaces employing up to 1000 individuals.

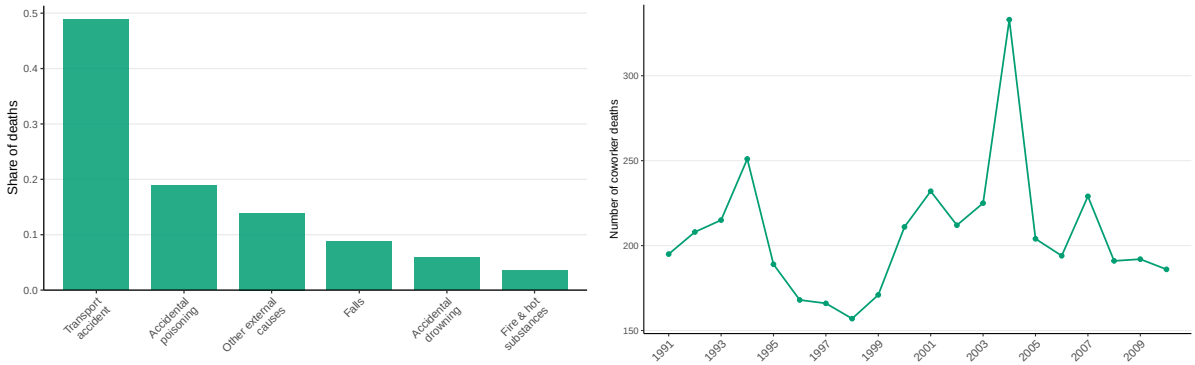
[9]Unexpected deaths are those due to transport accidents, accidental drowning, accidental poisoning, falls, fire and hot substances, and other external causes, corresponding to codes V01–V99, W65–W74, X40–X49, W00–W19, X00–X19, and W20–W64, W75–W99, X20–X39, or X50–X59 respectively in the ICD10 classification.

[10]2.3 per cent of treated individuals are affected by unexpected deaths in multiple years. Given the year-by-year matching procedure described below, individuals affected by multiple death events are treated as distinct observations and have a different set of past coworkers in each event.

Figure 1: Distribution of death events

(a) Cause of death

(b) Year of death



Notes: Distribution of the 4,129 included death events over causes and years. Numbers are calculated at the level of the deceased coworker (death event).

These correspond to two large civilian disasters.[11] I will therefore confirm that my main results are not sensitive to dropping any specific year from the sample. The unmatched sample also includes 650.7 million control dyads, made up of 36.7 million control individuals and 27.3 million placebo deceased past coworkers.

3.4 Matching procedure and control dyads

Even medically unexpected deaths may be predictable in a statistical sense: past earnings trajectory, gender, education or age may be associated with the risk of exposure to the unexpected death of a past coworker. Indeed, simply having a larger set of connections implies a greater risk of one of those connections dying unexpectedly. To ensure that such associations do not bias the comparison of treated and control individuals, I employ a matching strategy, similar to Azoulay et al. (2010) and Jäger and Heining (2022). First, I form strata based on the employment status in $t-1$ of the focal individual and their past coworker and, when the past coworker is employed in $t-1$, an indicator for whether the connection they provide to their current workplace is unique or not.[12] Second, I match treated-deceased dyads within strata without replacement to a set of ten control-placebo dyads, using an estimated propensity score.

The set of predictors used to estimate the propensity score is described in detail in Section B.1

[11] The 2004 Indian Ocean earthquake and tsunami and the sinking of the MS *Estonia* in 1994.

[12] A connection is unique if the focal worker has no other past coworkers employed at the same workplace.

of the Online appendix. The predictors include economic and demographic characteristics of both the deceased individual and the focal individual, as well as characteristics of the dyad and the topology of the network around each individual. Because of the rich nature of the administrative data, there are potentially many predictors to consider.[13] Furthermore, treatment is rare, raising the risk of overfitting in the propensity score model.

Rather than make ad hoc decisions about which predictors to include, I use a machine learning algorithm, the post-double selection procedure proposed by Belloni et al. (2014) (see also Fluchtmann et al., 2024, for a recent application to propensity score estimation). The algorithm selects strong predictors of exposure to an unexpected death as well as of the outcome (the change in treated individuals' earnings between $t-1$ and $t+2$). It then uses the union of the two sets of selected predictors to estimate a propensity score for the probability that a connection is lost due to an unexpected death. I use post-double selection rather than, say, a single LASSO as there may be important predictors of outcomes that are poor predictors of treatment status (e.g. the focal individual's past earnings). The post-double selection procedure ensures that such variables are included in the propensity score estimation, improving the efficiency of the matched estimator. I repeat the procedure and estimate a separate propensity score in each year, to account for variation over time in the distribution of causes of death as well as business cycle effects. The details of the propensity score estimation are in Section B.2 of the Online appendix.

3.5 Analysis sample

I report summary statistics on the unmatched and matched samples in Table C.1. There are only small differences in the topology of the network around treated and unmatched control dyads or in the strength of the connection. Larger differences appear between treated and (unmatched) control focal individuals. Treated individuals are on average more likely to be male and to not have completed high school, have five fewer employed contacts and are connected to three fewer workplaces. They are also slightly more likely to have experienced the death of a past coworker in the pre-treatment period. Deceased past coworkers are slightly older than unmatched placebo

[13] Including quadratic terms, indicators for categorical variables, and interactions between variables, the set of potential predictors contains 6721 variables.

past coworkers, have lower levels of education, and are much more likely to be male. The largest differences, however, are in the pre-treatment economic outcomes of past coworkers. Deceased past coworkers are 17 percentage points less likely to be employed in $t-1$ than unmatched placebo past coworkers and in general have lower earnings and employment throughout the network-building phase. Lower employment leads to smaller networks; deceased individuals have 10 fewer employed contacts and are connected to seven fewer workplaces than unmatched placebos. In general, the matching procedure leads to good balance on the matching variables, with no economically meaningful imbalances between the two groups.[14]

I report the one-digit industry in which focal individual and past coworker last worked together in Figure C.2. In general, treated dyads are drawn from a broad set of industries and are not concentrated in one or two particularly dangerous industries, such as Mining and quarrying. Treated dyads are more likely to have been formed in Food and beverage manufacturing, Construction, and Land transport and transport via pipelines than unmatched controls and are less likely to have been formed in Wholesale and retail trade, Education, or Health and social work. This is consistent with gender the educational differences previously observed between deceased and unmatched placebo past coworkers. Again, no economically meaningful imbalances remain after matching.

4 The average value of a professional connection

4.1 Event study design

4.1.1 Estimation

To estimate the average effect of the death of a past coworker on earnings and employment, I conduct an event study on the matched sample of treated and control individuals. Matched control difference-in-difference estimators have become the standard approach to estimating the effect of death as a treatment (e.g. Azoulay et al., 2010; van den Berg et al., 2017; Jäger and

[14] Some small differences persist between deceased past coworkers and matched placebo past coworkers, indicating that these covariates are not strong predictors of a dyad's treatment status. Consistent with this interpretation, these past coworker characteristics are not frequently selected by the post-double selection algorithm, see Table C.2. Furthermore, imbalances in treated and control dyads' characteristics do not affect the identification of the average effect of the death of a coworker, in Section 4, which relies on a parallel trends assumption.

Heining, 2022; Jensen and Zhang, 2026). The general empirical specification is summarised by the following equation:

$$y_{ijtl} = \alpha_{ijt} + \sum_{s=-5}^{6} \gamma_s \mathbf{1}(l = s) + \sum_{s \neq -1, s=-5}^{6} \beta_s \mathbf{1}(l = s, ijt = treated) + \epsilon_{il}. \tag{2}$$

The outcome of interest (earnings or employment) is denoted y_{ijtl} for focal individual i, who is exposed to the death of past coworker j in calendar year t. The outcome is observed in event year l, where $l = 0$ corresponds to the year of death, t. Individuals affected by multiple events in different years are considered distinct observations; ijt therefore indexes unique panel observations. α_{ijt} is an individual fixed effect. γ_s measure the effect of event time dummies, while β_s measure the dynamic treatment effect.

The panel is stacked, so it is balanced in event time; individuals not observed in the administrative registers in a given year are recorded as having zero earnings. Since the panel is balanced and adoption is not staggered in event time, Equation (2) can be estimated as a two-way fixed effects regression. Two-way fixed effects is, in this case, numerically equivalent to the difference-in-difference estimator of Callaway and Sant'Anna (2021). Treatment is defined at the level of the deceased or placebo individual j; I therefore cluster standard errors by past coworker by event year.

4.1.2 Discussion

The matched event study specification identifies the reduced-form effect of the death of a past coworker on subsequent outcomes under a parallel trends assumption. Specifically, if the untreated potential outcomes follow a parallel trend across the treated and matched control individuals, then the coefficients on the post-treatment dummies β_s, $s \geq 0$, identify the ATT, the average effect of unexpectedly losing a past coworker on the treated.

I show in Table C.1 that there are no detectable pre-trends in treated individuals' earnings or employment rates, relative to matched control individuals. However, the parallel trends assumption could still be violated if there is autocorrelation in focal individuals' exposure to past coworker deaths. To explore this possibility, I plot year-by-year differences in total all-cause mortality between treated and control individuals' past coworkers in Figure C.3. The set

of past coworkers is defined in year $t-1$ and held constant.[15] By construction, this difference is one in the event year, t. There is otherwise no evidence of either positive autocorrelation in exposure to past coworker mortality, which could arise if treated individuals' past coworkers faced a higher latent risk of mortality, or negative autocorrelation, which could arise if there were mean-reversion in exposure to past coworker mortality. Instead, the death of a past coworker leads to a one-off reduction in the size of a treated individual's professional network.

The theoretical parameter of interest is the l period-ahead effect of losing a connection, τ_{ik}^{l}, defined in Section 2. The reduced-form event study treatment effects β_s identify this parameter for $s = l$ under the condition that the death of a past coworker affects earnings and employment only through its effect on the size and composition of the job-search network. This exclusion restriction holds if workers do not otherwise change their behaviour following the death of a past coworker, e.g. by voluntarily working fewer hours or exiting the labour force. I will discuss the plausibility of the exclusion restriction at various points throughout the presentation of the results, in particular in Section 4.3. However, the maximum age restriction imposed on the treated individuals implies that early retirement is not an option following the death of a past coworker, given pension eligibility requirements.

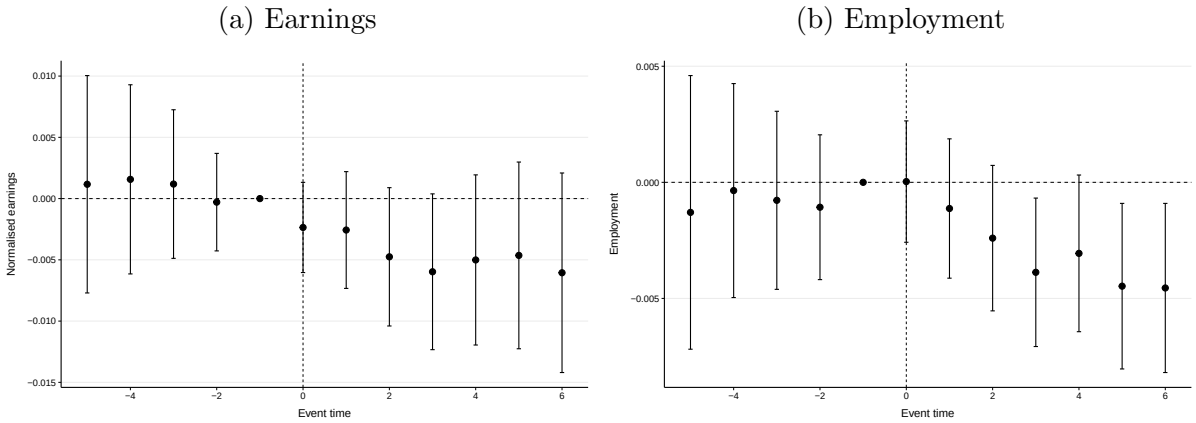
4.2 Average effect on earnings and employment

The average dynamic effect of the unexpected death of a past coworker on labour earnings is shown in Figure 2a. Affected individuals' earnings decline relative to the matched controls, however the effect is not immediate. Two or more years after the death, affected individuals' earnings are 0.5–0.6 per cent lower than they would have been, or approximately 800–1200 SEK. This effect is only significant at the 10 per cent level in years $t+2$ and $t+3$. In later years, the effect is approximately constant, however standard errors become larger. Pre-treatment earnings of the treated individuals follows the same trend as the matched control individuals, as seen previously.

The average dynamic effect of the unexpected death of a past coworker on employment is shown in Figure 2b. The time pattern of effects is very similar to what was observed for earnings.

[15] Mortality for either treated or control individuals' past coworkers may therefore be positive for $l < 0$, since I do not impose the restriction that all past coworkers be alive in $t-1$.

Figure 2: Labour market effect of unexpected past coworker deaths



Note: The figures report event-study estimates the effect of unexpected deaths of past coworkers, occurring during 1991–2010. Real annual labour earnings, measured in 2018 Swedish crowns, are normalised by average earnings in the control group in each event year by death year (Panel A). Employment is a dummy for having positive labour earnings during the calendar year (Panel B). Coefficients are estimated on the matched sample, using Equation (2). Standard errors are clustered by past coworker by event year, 95 per cent confidence intervals are reported.

Employment of affected individuals gradually declines, relative to the control group. The effect becomes significant at the five per cent level three years after the event, when the treated are 0.4 percentage points less likely to be employed. The reduction in employment is relatively stable after three years, at 0.4–0.45 percentage-points, or a 0.45–0.5 per cent. Approximately 65 to 80 per cent of the earnings effect can therefore be attributed to a decline in employment.

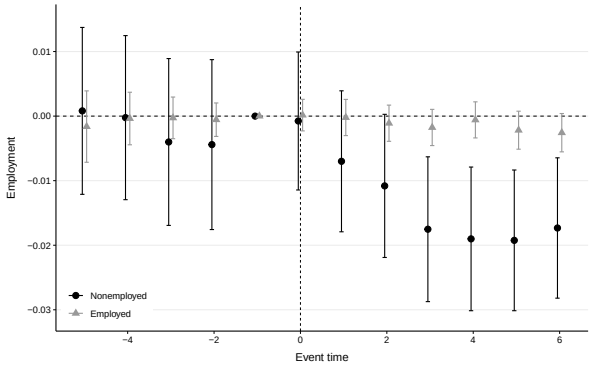
4.3 Mechanisms and access to jobs

The unexpected death of a past coworker could in principle reduce employment and earnings through multiple channels. Here, I provide evidence that a reduced flow of information about jobs from the treated individual's professional network is likely to be the main mechanism explaining the estimated earnings and employment effects.

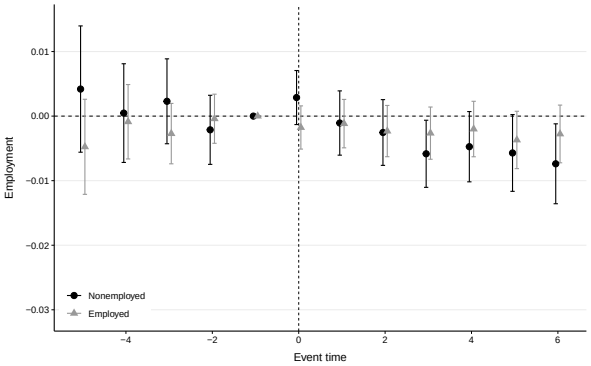
In Panels (a) and (b) of Figure 3, I first explore how the estimated employment effects vary with the employment status of the focal individual and their past coworker. Consistent with the predictions of the model in Section 2, I find large, negative employment effects for individuals who were nonemployed in $t-1$. They are nearly two percentage points less likely to be employed in year $t+3$, while the effect is small and insignificant for previously employed individuals. On the other hand, I find no significant differences in treatment effects by the

Figure 3: Treatment and access to jobs

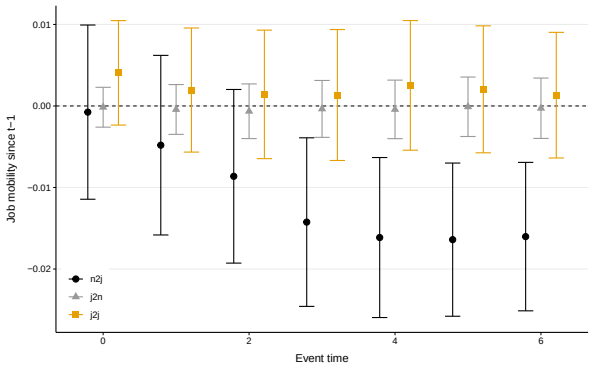
(a) Focal individual's employment, $t-1$



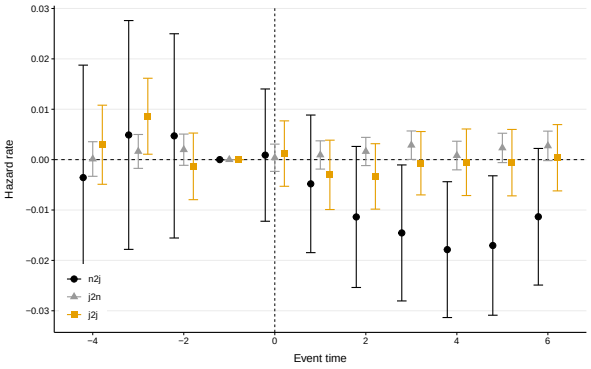
(b) Past coworker's employment, $t-1$



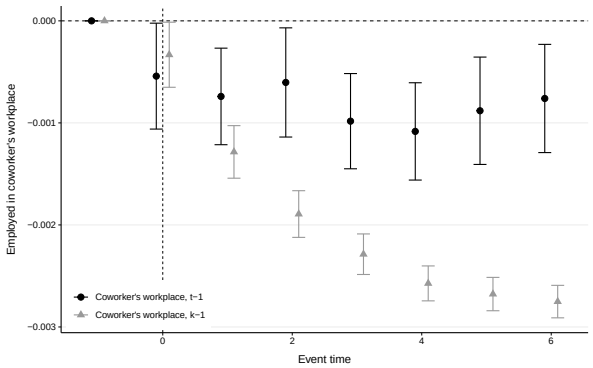
(c) Job transitions since $t-1$



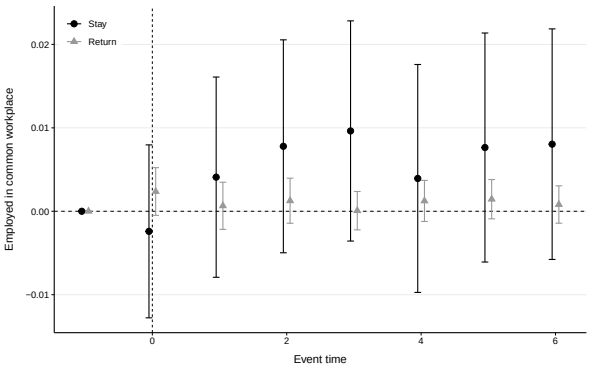
(d) Job transition hazard rates



(e) Referrals from past coworker



(f) Employment in past workplace



Notes: Panels (a) and (b) report difference-in-difference estimates, splitting the full sample by either the focal worker or the deceased coworker's employment status in $t-1$. Panel (c) reports simple differences between treated and control individuals, splitting the full sample by employment status in $t-1$; the outcome is an indicator for at least one transition relative to status in $t-1$ and is an absorbing state. Panel (d) reports difference-in-difference estimates, splitting the full sample by employment status in the event year l; the outcome is an indicator for a transition between l and $l+1$. Panel (e) reports simple differences between treated and control individuals. The outcome is either an indicator for having joined the deceased's workplace in $t-1$ (for the sample of individuals whose coworker was employed in $t-1$) of an indicator for being in an employment spell that started since $t-1$ and where the past coworker was employed in the year before the start of the spell(for the full sample). Panel (f) reports simple differences between treated and control individuals, splitting the full sample on an indicator for having left the common workplace in $t-1$. Standard errors are clustered by past coworker and year of death and 95 per cent confidence intervals are reported in all cases.

prior employment status of the deceased coworker. The employment status of the deceased worker was predicted in Section 2 to have an weak relationship with the sign of the treatment effect. Specifically, while employed connections are predicted to be weakly more valuable than unemployed connections, this value decreases in the number of unemployed connections they have, due to competition for information. The lack of any significant medium-term heterogeneity by the coworker's employment status suggests that competition for information may dominate.

Two caveats apply to this exercise. First, I only measure employment at an annual frequency, leading to measurement error in both workers' employment statuses at the time of treatment. This may attenuate the underlying patterns of heterogeneity. Second, the majority of nonemployed connections would soon have re-entered employment, had they survived, as shown in Figure C.4. Losing a nonemployed contact may therefore reduce access to job information in the medium run, even when there is competition for information, further reducing the scope for heterogeneity along this dimension.[16]

Further analysis of the effect of treatment on job transitions, in Panels (c) and (d) of Figure 3, confirms that the treatment mainly reduces the probability of leaving unemployment, rather than entering it. In Panel (c), I report differences in nonemployment-to-job, job-to-nonemployment, and job-to-job transitions. Focal workers' initial state is fixed in period $t-1$ and I treat a completed transition as an absorbing state. Treated nonemployed workers are less likely to transition into employment. After three years they are 1.5 percentage points more likely to be in the same nonemployment spell than control workers. Treated employed workers, on the other hand, experience no effect on the the likelihood of losing their job. They appear somewhat more likely to leave their ongoing job for another job, though these effects are small and not significant.

In Panel (d), I report difference-in-difference estimates of the relationship between treatment and the probability of exiting an ongoing spell, including all ongoing spells in a given event year l.[17] The negative association between the treatment and the probability of exiting

[16] I show in Appendix Section A that the conceptual framework predicts that the effect in period $l = 0$ for unemployed focal individuals is positive if they lose an unemployed connection, negative otherwise. Because individual measurement error in employment status at the time of treatment compounds at the dyad level, it is not possible to test this sharp prediction with the data available. For example, if 25 per cent of individuals' employment status is measured with error, 43 per cent of dyads' employment status will be measured with error.

[17] The results in panel (d) should not be interpreted causally, as the composition of the employed and

nonemployment declines after four or more years. This pattern implies that hazard rates for unemployment spells that start four or more years after treatment, by which time the treated individual's network has been partially renewed, are at most weakly affected by the loss of a former coworker. There is again little evidence of either job-to-job or job-to-nonemployment hazard rates being affected by the treatment.

Next, I study how likely the treated individual would have been to end up working with the deceased former coworker, had they not died. In Panel (e) of Figure 3, I plot treatment-control differences in the probability of working in the coworker's last workplace, defined in $t-1$.[18] Treated individuals are 0.1 percentage points less likely to be employed in this workplace three years after the event. In the counterfactual where a past coworker had not died, they would probably not have stayed in the same workplace throughout the post-treatment period. I therefore also report treatment-control differences for an indicator that is equal to one when the focal individual's ongoing employment spell (i) started after year $t-1$; and (ii) is in a workplace that employed the coworker in the year before the start of the spell. This type of connected employment is normally interpreted as indicating that the past coworker referred the focal individual to their employer at the time of hiring (Glitz and Vejlin, 2021). Here I find larger effects; treated individuals are 0.25 percentage points less likely to have received a referral from the deceased worker into their current workplace, three years after the event. While this effect is lower in magnitude than the full employment effect, it nevertheless suggests that the majority of lost information about vacancies relates to vacancies in the past coworker's current workplace. The loss of information about jobs in other firms, for example firms the past coworker is themselves connected to (i.e. the focal worker's second-degree connections) must therefore account for a smaller share of the observed employment effect.

Finally, I study whether the treatment makes workers more or less attached to their past workplaces, and in particular the workplace they shared with the deceased coworker. In Panel (f) of Figure 3, I report treatment-control differences in the probability of working in the last shared workplace, differentiating between focal individuals who are still in the workplace and ones who have left.[19] I do not find any evidence that treatment makes workers more or less likely

non-employed groups may change endogenously following the treatment.

[18] The sample is therefore restricted to observations where the past coworker was employed in $t-1$.

[19] The past coworker has always left the last common workplace, by construction.

to return to the workplace they shared with the deceased individual, consistent with the more general finding that job-to-job transitions are not affected by the treatment (see Panels (c) and (d)). There is some, albeit weak, evidence that treated individuals who had not yet left the past workplace by $t-1$ become less likely to do so post-treatment, consistent with reduced information about vacancies. However, these estimates are imprecise and not significant, reflecting the fact that few focal workers continue to be employed in the common workplace at the time of the event.

While the evidence presented here points to reduced access to information about vacancies being the main mechanism through which the estimated treatment effect operates, it does not by itself rule out other potential mechanisms. Data that could be used to evaluate competing mechanisms is somewhat limited.[20] Nevertheless, I explore evidence for two alternative mechanisms. First, I consider the evidence for labour demand effects. Deceased individuals typically have firm-specific human capital that may be hard to replace (Jäger and Heining, 2022). If the deceased individual's employer cannot replace them, their workplace may shut down, reducing demand for the focal individual's labour. I do indeed find that treated workplaces are 6.5 percentage points more likely to have shut down by $t+3$ than control workplaces, as shown in Figure C.5a. However, the workplaces induced by the treatment into closing are very small, as shown in Figure C.5b. The median control workplace that shuts employed 16 individuals in $t-1$, while the median treated workplace that closes employed 9 individuals. The closures caused by the treatment should therefore not be interpreted as aggregate labour demand shocks. Instead, they are a negative shock to precisely the focal individual's idiosyncratic labour demand curve, given their professional network.

Second, I explore the effect of the treatment on the focal worker's own mortality. Mortality, whatever the cause, may be symptomatic of more extreme cases of distress following the death of a close acquaintance, which is known to worsen individuals' labour market outcomes (van den

[20] In particular, I lack data on antidepressant and other drug use which has been used to show that grief and psychological distress following the death of a very close relative are important mechanisms for explaining the labour market effects of these events (van den Berg et al., 2017; Jensen and Zhang, 2026). Furthermore, I do not have data on family ties, to test whether the deceased coworker was a close relative of the treated individual and thus a possible source of nepotistic favours (Wang, 2013). In general, however, the unexpected nature of the treatment, and the fact that exposure to the treatment arises through past coemployment, make it unlikely that close family ties exist between the focal individual and the past coworker.

Berg et al., 2017; Jensen and Zhang, 2026). However, job loss is also known to increase mortality (Sullivan and Von Wachter, 2009) and income and mortality are in general closely associated (Chetty et al., 2016). I report the effect of an unexpected past coworker death on own mortality in Figure C.6. There is a precisely estimated zero effect of treatment on own mortality in the first years after the event. There is a small positive effect from year $l = 3$, by which time the treated individuals are 0.04 percentage points more likely to have died, increasing in magnitude and become significant by year $l = 6$.

The time pattern of mortality effects is arguably more consistent with increased unemployment in the short term raising mortality in the medium term than with grief as a mediator, as grief is experienced immediately after the death of a close acquaintance (van den Berg et al., 2017; Jensen and Zhang, 2026). However, the data available do not allow us to conclusively reject one or other explanation. Given the small effect sizes, less than 10 per cent of the employment effect, we can nevertheless rule out that death itself is a mechanism for the main employment effects. I confirm in Figure C.7 that restricting the sample to the endogenously selected group of individuals who survive until $t + 6$ has not effect on the average labour market effects.

4.4 Additional results

I conclude this section by discussing the robustness of the main event study results to different choices of methods and data. The post-double selection matching procedure uses a large set of covariates to achieve balance. I obtain very similar event study estimates if I instead perform random matching within cells constructed using a smaller number of covariates (Figure C.8).[21]

As noted previously, the number of deaths and the underlying distribution of causes of death vary from year to year. I therefore repeat my main matched event study specification, focusing on effects at $t + 3$, dropping each calender year in turn from the sample (Figure C.9). There is no evidence of the average earnings or employment effects being driven by coworker deaths in any specific year.

[21] I define cells using the focal worker's employment rate in the network building phase, $[t - 6, t - 2]$, dummies for their employment in $t - 1$, having above-median earnings in $t - 1$, sex, having completed less than a high school education, and the same covariates for the past coworker as well as the past coworker's earnings decile in $t - 1$ and 30 categories for past coworkers' own number of past coworkers. I randomly select three matches within the cell for each treated dyad, discarding dyads were there are fewer than three possible matches. I find matches for 96,513 treated dyads, or 98.9 per cent of the main analysis sample.

5 Heterogeneous treatment effects

In Section 4.3, I explored heterogeneous treatment effects by prior employment status, in order to understand how exposure to past coworker deaths affects transitions into and out of employment. However, an extensive theoretical and empirical literature has identified many potential sources of heterogeneity in the value of a professional connection. In this section, I explore how the effect of losing a past coworker varies with local network topology, demographic and socioeconomic characteristics of the focal individual and their past coworker, or homophily between the dyad members.

5.1 The Generalised Random Forest

5.1.1 Estimation

To identify covariates that predict heterogeneous treatment effects, I implement a machine learning algorithm, the generalised random forest (GRF, Athey et al., 2019), on the sample of treated dyads and matched controls. The GRF is an alternative to traditional heterogeneity analysis, which would involve estimating versions of Equation (2) where the treatment is interacted with covariates thought to predict heterogeneous treatment effects. Because there is a large number of theoretically relevant covariates, the classical approach may lead to overfitting, identifying spurious associations between covariates and treatment effect size. The classical approach is particularly ill-suited to studying heterogeneity if there are interactions between individual and network characteristics or if the relationship between treatment effect and some characteristics is non-linear.

The GRF estimates a series of causal trees (Athey and Imbens, 2016); a tree is a series of splits of the data. At each split the algorithm chooses a covariate and a value of the covariate on which to split so as to maximise treatment effect heterogeneity between the subsets of the data created by the split. Trees flexibly handle potential interactions and non-linearities in the relationship between the different covariates. The causal effect for each terminal node of the tree, or leaf, is estimated as the average augmented inverse propensity weighted (AIPW) score for observations that fall in that leaf. The output of each tree is a predicted conditional average treatment effect (CATE), given an observation's covariates. The GRF averages over trees to

generate a predicted CATE for each observation.[22]

To implement the GRF, I use the change in employment status between $t+3$ and $t-1$ as an outcome. I focus on changes, rather than levels, to link the GRF estimates to the event study estimates presented in Section 4.2, as I discuss below. I focus on employment, rather than earnings, for both conceptual and practical reasons. First, the results in Section 4 suggests that most of the estimated earnings decline is due to reduced employment, and in particular reduced job-finding. Second, focusing on employment reduces the risk that heterogeneity in estimated treatment effects partly reflects heterogeneity in human capital, which causes differences in baseline earnings growth. Third, when the variation in individual treatment effects is small relative to variation in the outcome under no-treatment, more trees are required to avoid overstating the degree of heterogeneity in treatment effects, increasing the computational burden. The standard deviation of $y_{t+3} - y_{t-1}$ in the control group is 0.60 for normalised earnings and 0.39 for employment, suggesting a lower risk of over-dispersion in CATEs for changes in employment. The set of predictors considered is the same as those used to estimate the propensity score, described in Section B.1 of the Online appendix.[23]

To estimate the GRF, I randomly split the data into five folds. I leave out one fold of the data at a time and fit the GRF on the remaining folds. For each fold, I use the estimated GRF to predict the CATEs in the left-out data. Each sampling step in fitting the GRF (splitting into folds, subsampling to train trees, and honest estimation) is *clustered*. I define clusters by partitioning the dyadic dataset within event years on both the focal individual and the past coworker and then taking the join of both partitions.[24] This way, no information on either individual is used in any of the steps to construct the CATE for a dyad. Sampling clusters, rather than dyads, ensures that information about individuals affected by the same event cannot leak across subsamples, which could lead to overfitting.

[22] To avoid overfitting, the GRF relies on random subsampling. Each tree is estimated on a random subset of the data and of the covariates. This random subset is then further split in two subsets, one for identifying the splits and another for estimating causal effects in each leaf, a procedure known as *honesty*. The CATEs produced by the GRF for each observation are an average over trees that did not include that observation in the training data.

[23] I do not include interactions and higher order terms, as the GRF flexibly considers interactions and non-linearities. The final set of predictors includes 319 variables.

[24] For treated dyads, the join simply corresponds to the partition on past coworkers. For controls dyads, both focal individual and past coworker can enter the sample multiple times in an event year, as part of different matched control dyads.

5.1.2 Discussion

The GRF produces valid causal estimates under two assumptions. The first is treatment *unconfoundedness* (Imbens and Rubin, 2015). Conditional on the included covariates, treatment assignment must be as good as random. The unexpected nature of former coworker deaths provides strong support for treatment unconfoundedness, even unconditionally, in the matched sample.

Because I use changes in employment as the outcome for training the GRF, the unconfoundedness assumption is closely related to the parallel trends assumption made in Section 4. Let $\Delta Y_i(T_i) \equiv Y_{i1}(T_i) - Y_{i0}$ denote the counterfactual change in the outcome between periods 0 and 1 for an individual with treatment status T_i. First, note that conditional unconfoundedness in the change in outcomes, $\big(\Delta Y_i(0), \Delta Y_i(1)\big) \perp T_i | X_i$, directly implies conditional parallel trends in the outcome $\mathrm{E}[Y_{i1}(0) - Y_{i0} | T_i = 1, X_i] = \mathrm{E}[Y_{i1}(0) - Y_{i0} | T_i = 0, X_i]$. Second, if the covariates are balanced in the sense that $X_i \perp T_i$, the treatment is *unconditionally* unconfounded with respect to the potential outcomes, $\big(\Delta Y_i(0), \Delta Y_i(1)\big) \perp T_i$. Unconditional unconfoundedness in the sample directly implies the unconditional parallel trends assumption made in Section 4. Table C.1 provides strong support for the balance condition. Conditional and unconditional parallel trends therefore coincide in this setting and conditional unconfoundedness is a stronger assumption than parallel trends.[25]

The second assumption the GRF relies on is treatment propensity *overlap* between treated and control individuals (Imbens and Rubin, 2015), given that causal effects in each leaf are estimated via AIPW. This assumption would almost certainly be satisfied if I re-used the estimated propensity on which treated and control individuals were matched, described in Section 3.4. However, those treatment propensities are estimated without splitting the data into the folds used to train the random forest, so they cannot be used to rank dyads on treatment effects, as I do below. I therefore estimate a new propensity score using a random forest on the matched sample, leaving out each fold one at a time.[26] The distribution of estimated treatment propen-

[25] Note that this is not true in general. If covariates are not balanced, conditional unconfoundedness is neither necessary nor sufficient for unconditional parallel trends. The two assumptions, while related, will in this case not be nested.

[26] Estimating AIPW treatment effects also requires an outcome model. I train a regression forest on the change in the outcome between $t-1$ and $t+3$ on the matched sample, again, fold-by-fold. As in the estimation of the GRF, all sampling is clustered by the join of the focal individual and past coworker

sities for treated and control dyads are nearly identical and share a common support, as shown in Figure C.10. The average treatment propensity of the treated dyads is 0.083, the average treatment propensity of the control dyads is 0.081.

5.2 Heterogeneity in the value of a lost connection

5.2.1 Estimated heterogeneous treatment effects

For each individual, the GRF estimates an individual-specific treatment effect, given covariates, a CATE. I report the estimated CATEs in Figure 4a. There is substantial dispersion in the estimated individual treatment effects. For an observation at the 25th percentile of the CATE distribution, losing a past coworker reduces employment by 0.6 percentage points after three years, while an observation at the 75th percentile experiences an *increase* in employment of 0.1 percentage points. The average estimated effect across dyads of losing a connection is a 0.3 percentage point decline in employment. However, the calibration test proposed by Chernozhukov et al. (2025) suggests that the fitted GRF may be overstating the true degree of heterogeneity in individual treatment effects.[27] The test returns a coefficient of 0.6 (se = 0.24), somewhat less than one, though not statistically significantly so at the five per cent level, indicating that the estimated CATEs may be moderately over-dispersed.

To better understand the extent to which the GRF-estimated CATEs identify true heterogeneity in individual treatment effects, I perform the following validation exercise. Within each fold, I rank observations on predicted CATEs.[28] I then validate the ranking of individuals on CATEs by re-estimating the treatment effect within each quintile. I consider three estimates of the treatment effect within each quintile: the average of the estimated AIPW scores, which identifies the quintile ATE, a re-weighted average of the AIPW scores, which identifies the ATT, and the difference-in-differences estimator, which also identifies the ATT. All three estimates are very similar, though the average AIPW scores are more precise.

Three patterns stand out. First, the estimated treatment effects are non-decreasing across

and by event year partitions.

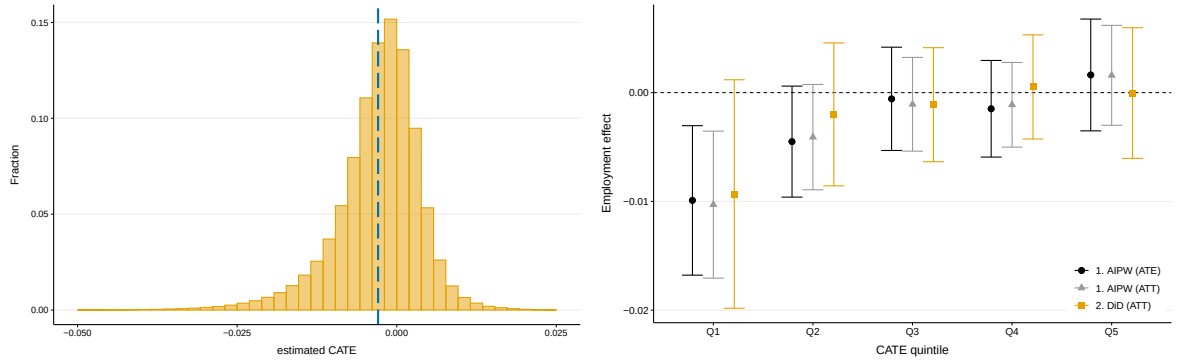
[27] To perform the test, regress the predicted AIPW scores on the demeaned CATEs. A slope coefficient of one would indicate that the GRF is well-calibrated, while a coefficient of zero would indicate that the identified heterogeneity in treatment effects is noise.

[28] Performing the ranking within folds ensures that data on neither dyad was used to fit the GRF when ordering the CATEs of two dyads.

Figure 4: GRF estimates of treatment effect heterogeneity

(a) Estimated CATE

(b) CATE quantile validation



Note: Distribution of estimated treatment effects (mean CATE = −0.003) and a validation exercise, re-estimating treatment effects within each quartile of estimated CATEs. The GRF is fitted using 10,000 trees and default parameter values. The outcome is the change in employment status between years $t-1$ and $t+3$. CATE quantiles are calculated within fold. For the validation exercise, 95 per cent confidence intervals, clustered by past coworker, are reported. For estimation details, see Section 5.1.1.

quintiles, suggesting that there is genuine heterogeneity in individual treatment effects and that the GRF identifies this heterogeneity. Second, negative treatment effects are concentrated at the bottom of the CATE distribution. Employment in the bottom quintile decreases by a whole percentage point, while in the second quintile, it decreases by a marginally insignificant 0.5 percentage points. Third, the estimated treatment effects are constant across quintiles 3–5 and close to zero. This confirms that the estimated CATEs are somewhat over-dispersed and that positive treatment effects at $t+3$ are unlikely.[29]

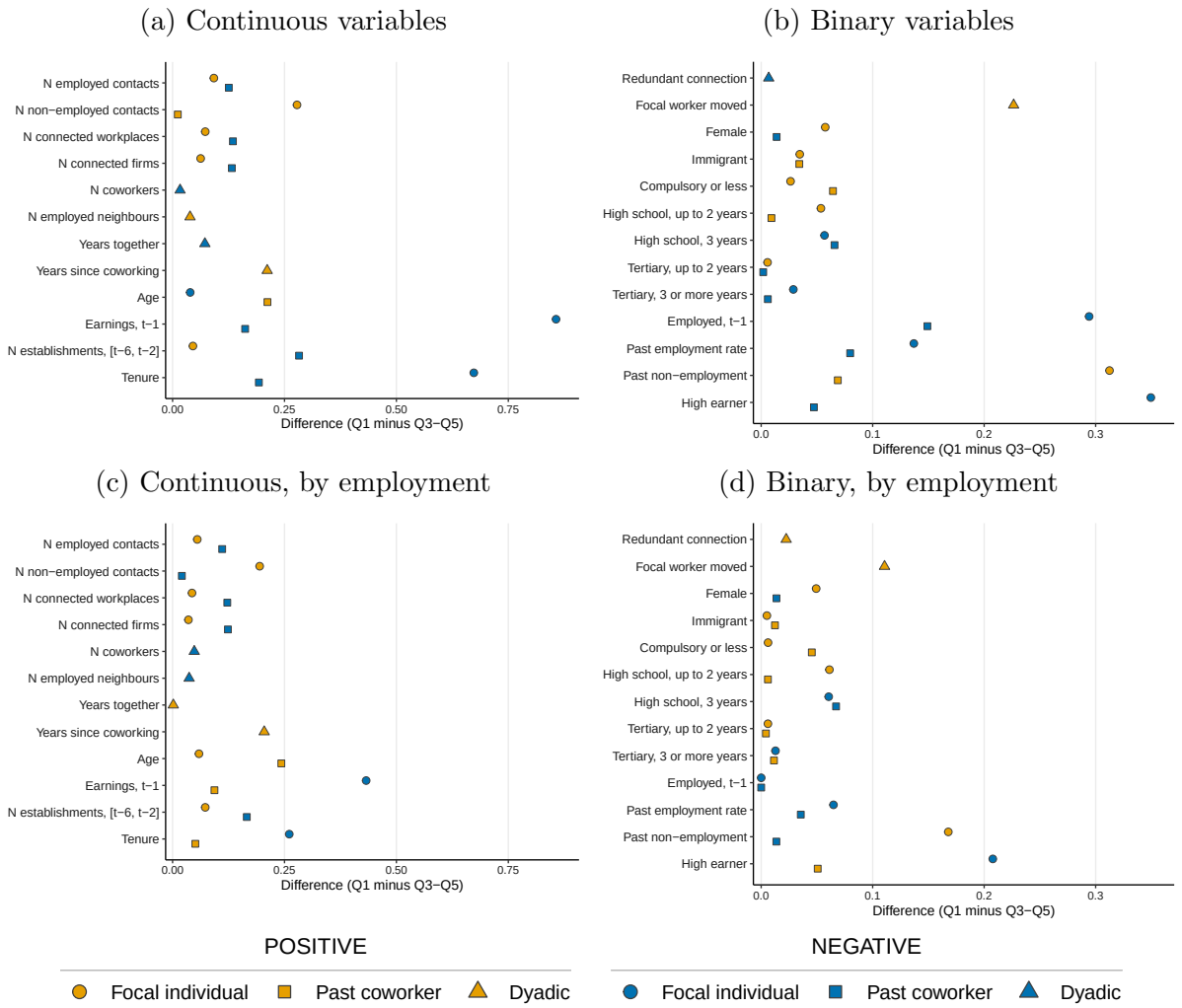
5.2.2 Predictors of heterogeneous effects

Having established that the GRF correctly identifies dyads where the focal individual experiences more negative employment effects from losing their past coworker, I now characterise predictors of negative treatment effects. Following Athey et al. (2026), I compute differences in the covariates used to train the GRF between dyads with large negative treatment effects (the first quintile of estimates CATEs) and dyads with a zero treatment effect (quintiles 3–5). I report differences for continuous variables, standardised to have mean 0 and standard deviation 1, in Figure 5a and differences for binary variables (and the employment rate during $[t-6, t-2]$,

[29]Increasing the number of trees in the GRF from 2000 to 10,000 only marginally increased the slope coefficient of the Chernozhukov et al. (2025) test, suggesting that any over-dispersion in the CATEs is not due to Monte-Carlo error in any of the sampling steps of the GRF.

measured on $[0, 1]$) in Figure 5b. These differences should not be interpreted causally. For example, while the average CATE conditional on employment in $t-1$ does identify the average causal effect of losing a connection for employed or unemployed individuals, the difference between these averages does not identify the causal effect of employment on the average CATE, as employment is correlated with other characteristics.[30]

Figure 5: Differences across CATE quantiles



Note: Absolute values of differences in average covariates between the bottom quintile of the predicted CATE distribution and quintiles 3–5. The sign of the difference and the unit on which the covariate is defined and measured is shown in the legend. Continuous covariates have been standardised to have mean 0 and standard deviation 1. Binary variables are either dummy variables or take values on $[0, 1]$. Quintiles are defined by ranking CATEs either within fold, or within fold and employment status in $t-1$ for both focal individual and past coworker.

Socioeconomic characteristics. The largest covariate differences between negative- and

[30] I do not report differences for the full set of 319 predictors used to train the GRF as many are weak predictors or are highly correlated with one another.

zero-effect dyads are in economic factors such as pre-treatment earnings and employment. For example, focal individuals in the bottom quintile of treatment effects are nearly 30 percentage points less likely to be employed in $t-1$, and their past coworkers are around 14 percentage points less likely to be employed. These differences are consistent with recent evidence unemployed workers' job search effort being complementary with their unemployed connections' search effort (Caria et al., 2023) and may reflect the direct effect of employment status at the time of treatment. However, these differences may also partly capture underlying differences in employability or attachment the labour market. I therefore re-rank all dyads within cells defined by GRF fold and the employment status in $t-1$ of both focal individual and past coworker. I re-compute covariate differences between the first and third to fifth quintiles of the within-employment status ranking, to identify predictors of negative treatment effects net of employment status. I report the results in Figures 5c and 5d and focus on these differences in the subsequent discussion.

Economic factors remain the most important predictor of treatment effect size, conditional on employment status in $t-1$. Focal individuals with the most negative treatment effects earn nearly half a standard deviation less in $t-1$, are more than 20 percentage points less likely have above-median earnings, and are around 17 percentage points more likely to have experienced a year of non-employment in $[t-6, t-2]$. On the other hand, the most valuable past coworkers, whose loss is associated with the most negative treatment effects, earn around 0.1 standard deviations more in $t-1$, are 5 percentage points more likely to have above-median earnings and and are 2 percentage points less likely to have experienced any non-employment during the network building phase. These findings are consistent with standard models of information sharing on networks (workers keep offers with wages above their own and pass on offers with lower wages to a connection who can use it, Calvó-Armengol and Jackson, 2004, 2007; Beaman, 2012) and with existing empirical evidence for these models (Cingano and Rosolia, 2012; Glitz, 2017).

Alternative measures of attachment to the labour market show a similar pattern of associations with treatment effects. Negatively affected focal individuals have more unstable employment during $[t-6, t-2]$, as measured by a higher likelihood of having moved on from the workplace where the connection was formed, a higher number of workplaces during the network

building phase, and a shorter tenure with their current employer in $t-1$. More valuable connections, on the other hand, have fewer employers during the network building phase and longer tenure in the year before treatment.

Other socio-demographic predictors of attachment to the labour market have similar effects on estimated treatment effects. Negatively affected focal individuals are 5 percentage points more likely to be women, notwithstanding existing evidence that women may be penalised by reliance on networks in job search in some contexts (Beaman et al., 2018; Cullen and Perez-Truglia, 2023; von Essen and Smith, 2023), while valuable connections are slightly more likely to be male, as Boucher and Goussé (2019) also find. Both negatively affected focal workers and valuable past coworkers are more likely to have not completed high school, consistent with existing findings that lower-educated workers use networks more intensively (Pellizzari, 2010; Kramarz and Skans, 2014). More valuable coworkers are older on average, while the age of the focal worker appears to matter less.

Network size and tie strength. According to standard models of job offer sharing on a network, the effect of a lost connection has an ambiguous relationship with the number of connections of both the focal individual and the past coworker.[31] Empirical evidence on the role of network structure is also mixed, with evidence both of stronger ties, with a higher degree of redundancy, mattering more (Gee et al., 2017a,b; Rajkumar et al., 2022) as well as evidence of crowding out in larger networks, reducing the value of a connection (Beaman, 2012; Eliason et al., 2023).

I find similarly mixed patterns. Negatively affected focal individuals have a larger network on average, whether measured using individual past coworkers (see also Appendix Figure C.11 for differences in total connections, irrespective of employment status in $t-1$) or connected workplaces and firms. More valuable connections on the other hand have smaller networks. Negative treatment effects are associated with fewer employed neighbours at the dyad level, but also with a higher probability of having more than one connection to the past coworker's current firm and a longer lag since both individuals worked together. There is, therefore, no straightfor-

[31] As discussed in Section 2, the precise relationship depends on the relative frequency of competition versus information effects, the employment status of connections, and the overlap between both individuals' networks. Under realistic parameter values, one might expect information effects to dominate and a negative association between either individual's network size and the effect of a lost connection.

ward relationship between network structure, tie strength, and the estimated treatment effects. However, these patterns should not be over-interpreted, as the measures of network size and tie strength I use here are strongly correlated with measures of labour market attachment such as number of past employers. I do nevertheless find that negatively affected individuals formed their connection in smaller workplaces, where focal individual and past coworker had 0.05 standard deviations fewer coworkers.

Homophily. In fitting the GRF, I do not include direct measures of homophily as predictors, instead allowing the algorithm to uncover any important interactions between focal and past coworker characteristics. To assess the role of homophily in predicting negative treatment effects, I construct a set of measures of socio-demographic and economic similarity between two individuals. I report differences in measured homophily between the bottom quintile of dyads with negative treatment effects and quintiles 3–5 of dyads where there are no earnings effects of losing a connection in Figure C.11. I find limited evidence of homophily predicting the effect of losing a connection, contrary to prior evidence on job information flowing more readily between similar individuals (McPherson et al., 2001; Pedulla and Pager, 2019). Focusing on differences across CATE quintiles defined within employment cells, I find that negatively affected focal workers are at most 1 percentage point more likely to share a given characteristics with their former coworker. In some cases, they are less likely to share a trait, in particular being male. On the other hand, I find that specific measures of *dissimilarity* are strong predictors of negative treatment effects. In particular, negatively affected focal individuals are 6 percentage points more likely to lose an older coworker, who was on average 8 ranks higher than them in the workplace earnings ranking, or 14 percentiles in the national earnings distribution, when they last worked together.

Taken together, the differences reported in Figures 5 and C.11 paint a coherent picture of when professional connections are most valuable. In particular, younger and less educated workers who are less strongly attached to the labour market are more likely to be negatively affected by the loss of a connection. More valuable connections are themselves older, higher earning, and more strongly attached to the labour market. Relating these findings to the theoretical framework presented in Section 2, the value of a connection appears to be primarily driven by differences across focal and past coworkers in job arrival and separation rates, a and b, rather

than differences in network structure or in the propensity to pass on information within a dyad, conditional on having a vacancy to share. In other words, the value of a connection is a function of needs more job information, and who can provide it. These findings are consistent with prior evidence that workers weakly attached to the labour market, such as the young, rely most on connections to find a job (Kramarz and Skans, 2014; Loury, 2006; Pellizzari, 2010). It also provides evidence on the external validity of research designs that focus on the networked job search of displaced workers (e.g. Cingano and Rosolia, 2012; Glitz, 2017; Eliason et al., 2023; Åslund et al., 2024).

Finally, the pattern of associations I document has implications for the interpretation of my definition of a connection. One possible explanation for the existence of heterogeneous treatment effects in my setting could be that my exposure-based measure of a connection captures both true connections, or friends, who pass on job information, as well as past coworkers with whom the focal worker formed no lasting bond, who do not pass on job information.[32] If this were the case, measures of gender, educational, or demographic similarity within the dyad, which predict with whom an individual is likely to form a tie, should be good predictors of negative treatment effects. Instead, my results suggest that treatment effect heterogeneity reflects heterogeneity in the demand for demand for information, on the focal worker side, and the supply of information, on the coworker side, rather than variation in the existence of a connection per se. I conclude that past workplace exposure is a good proxy for true connections, at least in the smaller workplaces I focus on.

5.2.3 Heterogeneity across and within deaths

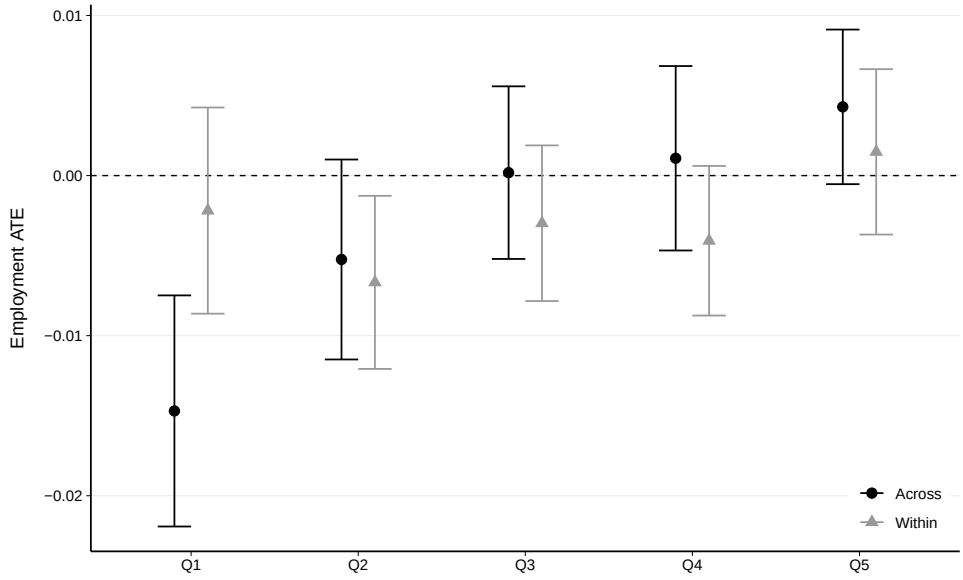
The heterogeneity analysis in the previous section showed that the characteristics of both the deceased and the treated individual are associated with negative treatment effects. To explore further how the characteristics of one or other member of a dyad matter, I decompose variation in treatment effects into variation *within* deceased individuals (i.e. across all treated individuals connected to the same deceased individual), and *across* deceased individuals.

To study variation within deceased individuals, I rank individuals affected by the same death on their estimated CATE and divide them into quintiles. I then pool rankings and estimate the

[32] See Chetty et al. (2022a,b) on the distinction between *exposure* and *friending*.

effect of a death as the average AIPW score for each within-deceased coworker quintile. To study variation in treatment effects across deceased individuals, for each focal individual I calculate the leave-out mean CATE of all other individuals affected by the same past coworker death. I pool observations and rank individuals within fold on the average CATE of their co-treated focal individuals. I again calculate the average effect of a death as the average AIPW scores within each quintile.[33]

Figure 6: Decomposition of variation in treatment effects



Note: Average AIPW scores for focal individuals ranked within deceased individual or across deceased individual CATE quartiles. Only deaths affecting at least five connected individuals who satisfy my sample restrictions are included. 95 per cent confidence intervals, clustered by deceased individual, are reported.

The results of the decomposition exercise are shown in Figure 6. There appears to be more variation in treatment effects across deceased individuals than across focal individuals affected by the same death. Again, negative effects are concentrated at the bottom of the CATE distribution. The fact that treatment effects vary more across deaths than within deaths has two possible interpretations. Either characteristics of the past coworker are the main determinant of the effect of losing a connection, or the characteristics of the focal individual that predict negative treatment effects are highly correlated within past coworker.

[33]I restrict attention to deceased coworkers connected to at least five individuals, which corresponds to 98 per cent of the main dyadic sample. Control individuals are assigned to the deceased coworker of their matched treated individual, as there are few cases where a matched placebo coworker is connected to multiple included matched control individuals.

To tease apart these two interpretations, I regress the dyad-level CATEs on a set of fixed effects for past coworkers. The adjusted R^2 of this regression is 0.07. I also regress the CATEs on the focal worker's employment status and a quadratic of their earnings in $t-1$, the two strongest predictors of the individual treatment effects. The adjusted R^2 of this regression is 0.14, while the R^2 of the regression including both the focal worker-level predictors and the past coworker fixed effects is 0.18.[34] These results suggest that the variation in treatment effects across deceased individuals is likely to reflect at least as much variation in the characteristics of the deceased individual as in the average characteristics of the treated focal individuals. However, the R^2 is relatively low in all cases, implying that more of the variation in treatment effects reflects variation in dyad-level characteristics.

6 Networks and inequality

6.1 The value of the marginal connection

Differences in social networks are widely thought to contribute to unequal labour market outcomes (Jackson, 2025). Professional networks may contribute to unequal earnings and employment rates if either the same connection is not equally productive in providing job information to different individuals, or if access to uniformly productive connections is unevenly distributed across worker types. The estimated CATEs for my sample dyads, presented in Section 5.2, can readily be used to study how the employment effect of access to a certain type of past coworker varies over individuals of different types. However, the fitted CATE function can also be used to understand how valuable connections are distributed across workers.

To understand how access to professional networks contributes to unequal job search outcomes, given both differences in the effectiveness of connections across types and differential access to types of connections, I predict the CATE for all controls workers' connections, not just those selected as matched placebo deceased individuals.[35] To understand how the value of

[34] In these regressions, I continue to pair control individuals with the past coworker of their matched treated individuals, for comparability with Figure 6. When I regress the CATEs on the (unpaired) set of past coworker fixed effects, the R^2 is 0.3 without including the focal worker-level predictors and 0.42 when including focal-worker predictors, as many placebo past coworkers are singletons.

[35] Conceptually, I exclude treated individuals as their surviving coworkers are not eligible placebo past coworkers in the original matching procedure. The focus on control workers only is without loss of

a worker's network varies by socio-demographic group, I report two measures of the value of a worker's network: the average predicted CATE across their connections, which can be interpreted as the expected reduction in employment at $t+3$ from removing a uniformly randomly chosen individual from a worker's network, and the minimum CATE across their connections, interpretable as the employment effect of losing their most valuable connection.

I report summaries of both these measures of network value by focal worker age and gender, nativity, or education, in Figure 7. The patterns of heterogeneity in the value of a lost connection are similar to what was found when focusing only on the deceased and matched placebo past coworkers. The average past coworker has little value for the selected sample of very young workers; from age 25 onwards the value of a worker's network is relatively stable. Women and immigrants have somewhat more valuable networks than men and and natives, a pattern that is particularly clear when focusing on the most valuable connection. Younger workers from the extremes of the educational distribution have more valuable networks, though as workers get older, the value of the network become inversely proportional to the amount of education a worker has. I conclude from these findings that differences in the employment effect of a connection documented in Section 5.2 are not undone by differences in the distribution of valuable connections across different types workers.

The predicted CATE is missing for some past coworkers, in particular those who previously worked in a large workplace, or a workplace with a missing identifier. I show in Appendix figure C.12 that the share of connections missing a predicted CATE does vary with age, rising in the first few years then falling from around age 25. However, this variation remains in a small range, between 36 and 42 per cent, and is therefore unlikely to explain the heterogeneity observed in Figure 7.

6.2 The total value of a network and inequality

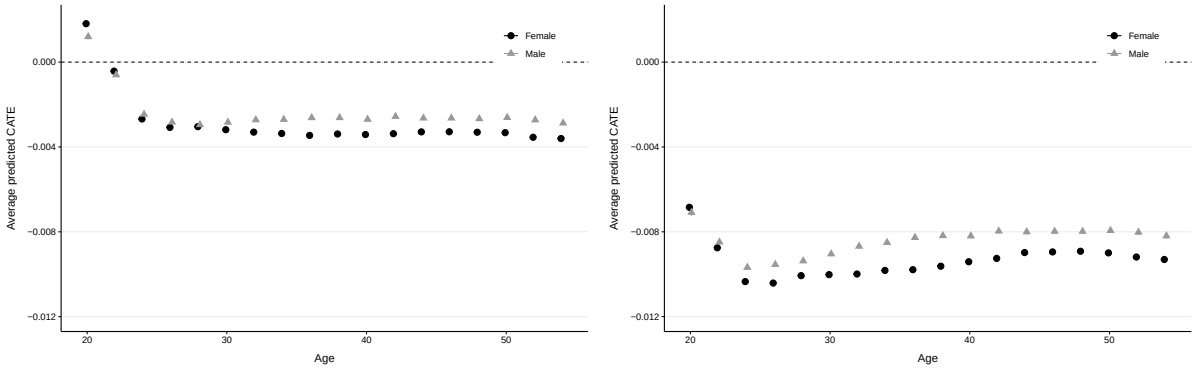
The findings in Sections 5.2 and 6.1 suggest that lower-earnings workers and socioeconomic groups are in general more harmed by the loss of a past coworker. However, the value of marginal connection is a function of the size and topology of the network (Section 5.2). If

generality, as treated and control workers are observationally identical on the covariates used to train the GRF, see Appendix Table C.1. Both groups will therefore have the same average CATEs.

Figure 7: The value of a worker's network

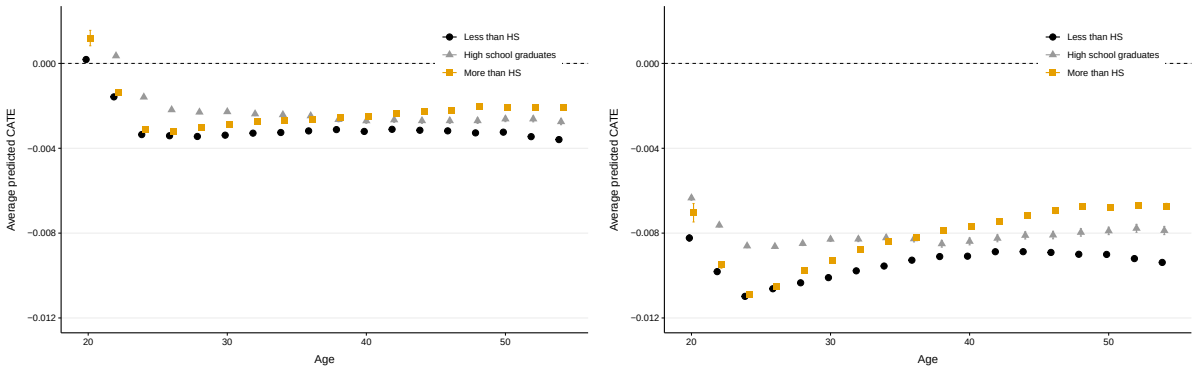
(a) Gender, average CATE

(b) Gender, minimum CATE



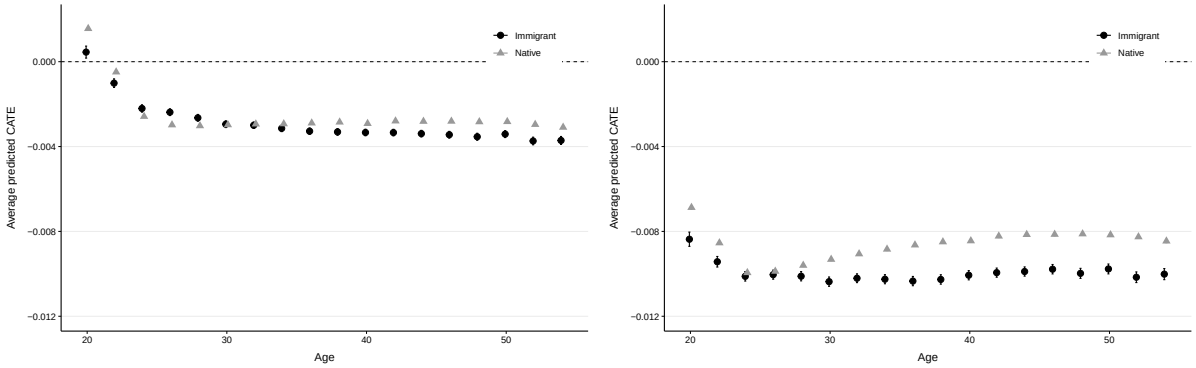
(c) Education, average CATE

(d) Education, minimum CATE



(e) Nativity, average CATE

(f) Nativity, minimum CATE



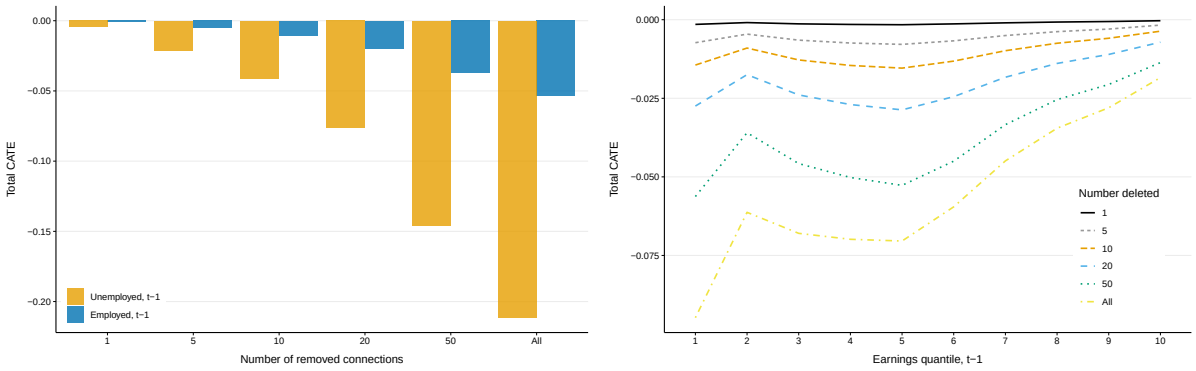
Note: Average and minimum CATEs by demographic characteristic. Age is grouped in two-year bins. Robust 95 per cent confidence intervals are reported.

network size is correlated with wages or employment, it is not possible to directly relate the average value of the marginal connection to inequality in earnings.

To use the CATE function fitted by the GRF to estimate the total value of different workers' networks, I use the following algorithm. At each step, I uniformly randomly select a connection for each focal individual to remove from their set of connections. I update the measures of network topology around each focal individual, given the position of their removed connection in the network.[36] I then re-predict the CATE for all remaining connections, given the new network structure. I repeat this procedure until all connections have been removed. The predicted CATEs are still highly correlated with the constant-network CATEs reported previously (0.88). However, the correlation does decline to 0.76 for coworkers who are removed after 50 or more iterations of the algorithm, underscoring that network structure may matter somewhat for the total value of a network.[37]

Figure 8: Total value of the network

(a) Employment status, $t-1$ (b) Earnings percentile, $t-1$



Note: In Figure A I report the total effect of removing a randomly drawn sequences of a worker's connections, differentiating by employment status in $t-1$. I update the network measures around a focal individual after each removal of a coworker. In Figure B I divide positive earnings in $t-1$ into deciles and report the total effect of a removing a sequence of connections, up to a given number of iterations of the algorithm, in each bin. Coworkers with a missing CATE are included in the sequence of removed coworkers, though their associated CATEs are still treated as missing when summing over coworkers.

[36] I update their number of employed and unemployed connections, connected firms and workplaces, their past coworkers' number of employed and unemployed connections, an indicator for these past coworkers providing a unique connection to their current employer, and the number of employed neighbours of the dyad, though not measures of past workplace size.

[37] This procedure, while informative, should be treated as providing only a rough guide to the employment effect of removing all an individual's connections. First, it does not account for the possibility that workers' search effort or use of different search technologies may respond differently to losing all connections than to losing a single connection. Second, shutting down the use of networks in job search in the whole economy, which this simulation speaks to, would have general equilibrium effects on vacancy posting and wages that the CATEs, derived from an event study, do not identify.

To relate the total value of workers' networks to inequality, I report the cumulative effect of iteratively removing up to 1, 5, 10, 20, 50, or all of a worker's connections, in a randomly drawn order, in Figure 8a, differentiating by employment status in $t-1$. Unemployed workers' networks are much more valuable, consistent with what was shown in Section 5.2. The total effect for an unemployed worker of losing access to their network is a 21 percentage point reduction in employment in $t+3$, while an employed worker's employment probability would decrease by only five percentage points if they lost access to their network. Furthermore, the effect of allowing network size to vary is at best modest. For a worker with at least 50 connections, the CATE of the first removed worker is on average -0.0021, while the CATE of the fiftieth removed worker is on average -0.0024.

Finally, I show how the total value of the network relates to labour earnings of employed individuals in $t-1$ in Figure 8b. Workers in the bottom decile of the earnings distribution are particularly affected by the loss of connections; their employment in $t+3$ would decline by 9.5 percentage points if they lost their entire network. Some of these workers may already have lost their job during year $t-1$, explaining the particularly large effects. The value of the network is then stable, and may even increase slightly over deciles 2–5, before decreasing rapidly as earnings go beyond the fifth decile. This broad pattern echoes the finding of Lester et al. (2026), who show that lower-skill workers are more dependent on offers from their network for finding work. However, even workers in the top wage decile would be negatively affected by the loss of their network; they would be 1.8 percentage points less likely to be employed in $t+3$ without their network. Nevertheless, these estimates clearly indicate that completely shutting down an individual's professional network would both (i) expose low-wage workers to a relatively larger unemployment risk than higher-wage workers; and (ii) make unemployment much more persistent. Both of these effects would lead to an increase in standard measures of inequality, implying that the use of professional connections in job search may reduce inequality.

7 Conclusion

In this paper I investigated the effect of losing a past coworker on an individual's earnings and employment. I found that the loss of a past coworker lowers annual earnings by an average of

0.6 per cent relative to control individuals (1,100 SEK) three years after the death and lowers the probability of employment by 0.4 percentage points. I showed that this effect is largely accounted for by a reduced flow of job offers to unemployed individuals.

The value of a lost connection is heterogeneous. Workers more weakly attached to the labour market are more harmed by the loss of a connection, while older, more economically successful connections, both absolutely and relative to the treated individual, are particularly valuable, as are connections with a small set of connections of their own, highlighting that job search information is rival. Because connections are more valuable for individuals who are more weakly connected to the labour market, the use of professional networks in job search may reduce labour market inequality on average, even after accounting for differences in the composition of networks.

A potential limitation of the findings presented here is that the study relies on relatively low-frequency data, with earnings measured annually, leading to error in the measurement of employment at the time of treatment. This measurement error may obscure changes in the amount of competition for information and jobs between unemployed individuals when a worker is removed from the network. In future research, the matched event study approach used here could be applied to unemployment spell data, to study the extent to which unemployed network members compete with each other. The fact that the earnings effect can mostly be explained by extensive margin employment effects may also be specific to the Swedish data used, as the Swedish wage distribution is compressed by international standards. Similar studies in other countries may find larger wage effects of losing a connection.

Finally, the fact that the use of professional connections may help to reduce earnings inequality does not imply that differential access to or productivity of connections formed in other settings, such as schools or neighbourhoods may not contribute to other forms of inequality. This may be particularly true when such connections affect important life decisions, such as whether to attend university. The approach developed in this paper, where I comprehensively study heterogeneous effects of quasi-experimental variation in exposure to some component of one's network, could productively be extended to other settings to understand how other layers of an individual's network may contribute to inequality at different points in the life-cycle.

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A Online appendix: Theoretical details

In this appendix, I derive the results referred to in the theoretical analysis of the effect of loosing a connection in Section 2.

A.1 Definitions

In any period t, the probability that an unemployed worker receives at least one job offer from their connections is

$$Q_{it} \equiv 1 - \prod_{\substack{j \neq i : g_{ij}=1 \\ s_{j,t-1}=1}} (1 - q_{jt}),$$

where the probability that employed worker j receives a job offer in t and passes it on to a given unemployed connection i is

$$q_{jt} \equiv \frac{a}{\sum_{j'=1}^{n} g_{jj'}(1 - s_{j't-1})}.$$

The probability that a worker i is employed at the end of t is

$$P(s_{it} = 1 | \mathbf{s}_{t-1}) = \begin{cases} 1 - b, & s_{it-1} = 1 \\ (1-b)\big[a + (1-a)Q_{it}\big], & s_{it-1} = 0. \end{cases}$$

Finally, define the *conditional* l period-ahead causal effect on i of removing connection k, conditional on a future realisation of the state vector:[1]

$$\kappa_{ik}^{l}(\mathbf{s}_{l-1}) \equiv P_{-k}(s_{il} = 1 | \mathbf{s}_{l-1}) - P(s_{il} = 1 | \mathbf{s}_{l-1}) \tag{A.1}$$

Partition i's set of connections into the subset that are also connections of k, I_k, and the subset that are not connections of k, I_{-k}. It follows directly from the definitions above that the

[1] The employment status of the removed connection, s_{kl-1}, is a counterfactual. It is a possible realisation of k's employment status in $l-1$, had they not been removed from the network. Because removing an individual from the network is equivalent to setting $g_{kj} = 0$ for all j, the transition probabilities under $P_{-k}(\cdot|\mathbf{s}_{l-1})$ are in any case invariant to the value of s_{kl-1}.

conditional causal effect defined in Equation (A.1) is:

$$\kappa_{ik}^{l}(\mathbf{s}_{l-1}) =$$

$$\begin{cases} 0, & s_{il-1} = 1 \\ -(1-b)(1-a)q_{kl}\prod_{\substack{j\neq k, j:g_{ij}=1 \\ s_{jl-1}=1}}(1-q_{jl}), & s_{il-1}=0, s_{kl-1}=1 \\ (1-b)(1-a)\prod_{\substack{j\in I_{-k} \\ s_{jl-1}=1}}(1-q_{jl})\left[\prod_{\substack{j\in I_k \\ s_{jl-1}=1}}(1-q_{jl}) - \prod_{\substack{j\in I_k \\ s_{jl-1}=1}}(1-q_{jl}^{-k})\right], & s_{il-1}=0, s_{kl-1}=0 \end{cases}$$

where q_{jl}^{-k} is the probability that j receives an offer and passes it to a given unemployed i in period l when we set $g_{jk} = 0$. I refer to the second case as the *information effect*. It measures the loss of information about vacancies that an employed k would have passed on to an unemployed i. By definition, $0 < q_{jl} < 1$, and $g_{ik} = 1, s_{il-1} = 0, s_{kl-1} = 1$ implies that $q_{kl} > 0$, so the information effect is negative. I refer to the third case as the *competition effect*. It measures the gain in information about vacancies for an unemployed i now that they are no longer competing with k to receive vacancies. It is straightforward to show that $q_{jl}^{-k} \geq q_{jl}$, so the competition effect is non-negative. Note also the following heterogeneity:

- **Information and number of employed connections:** adding an employed connection to i's set of connections, i.e. setting $g_{ij} = 1$ for some j where $s_{jl-1} = 1$ and leaving all other links $g_{ij'}$, $j \neq j'$, as before, will multiply the information effect by $0 < 1 - q_{jl} < 1$. The information effect is therefore weakly decreasing in absolute value in the number of i's employed connections.

- **Competition and number of employed connections:** adding an employed connection to I_{-k} will also multiply the competition effect by $0 < 1 - q_{jl} < 1$. In this case, the competition effect is weakly decreasing in absolute value in the number of i's employed connections in I_{-k}. When adding an employed connection to I_k, there is an additional change in the effect of removing k on the competition to receive a vacancy. The net effect of adding an employed connection in I_k on the competition effect is therefore ambiguous, though the competition effect will still be non-negative in all cases.

- **Number of unemployed connections:** q_{jl} is decreasing and convex in the number of

j's unemployed connections. The information effect is therefore decreasing in the number of k's unemployed connections and increasing in the number of unemployed connections of i's other employed connections, $j : g_{ij} = 1, s_{jl-1} = 1$. The competition effect is increasing in the number of unemployed connections of $j \in I_{-k}$ and decreasing in the number of unemployed connections of $j \in I_k$.

A.2 The l period-ahead causal effect

The unconditional l period-ahead causal effect, τ_{ik}^l, can be decomposed as follows:

$$\begin{aligned}
\tau_{ik}^l &= P_{-k}(s_{il} = 1) - P(s_{il} = 1) \\
&= \sum_{\mathbf{s}_{l-1}} \big[P_{-k}(s_{il} = 1|\mathbf{s}_{l-1})P_{-k}(\mathbf{s}_{l-1}) - P(s_{il} = 1|\mathbf{s}_{l-1})P(\mathbf{s}_{l-1})\big] \\
&= \sum_{\mathbf{s}_{l-1}} \big[P_{-k}(s_{il} = 1|\mathbf{s}_{l-1}) - P(s_{il} = 1|\mathbf{s}_{l-1})\big]P(\mathbf{s}_{l-1}) + \big[P_{-k}(\mathbf{s}_{l-1}) - P(\mathbf{s}_{l-1})\big]P_{-k}(s_{il} = 1|\mathbf{s}_{l-1}) \\
&= \sum_{\mathbf{s}_{l-1}} \kappa_{ik}^l P(\mathbf{s}_{l-1}) + \big[P_{-k}(\mathbf{s}_{l-1}) - P(\mathbf{s}_{l-1})\big]P_{-k}(s_{il} = 1|\mathbf{s}_{l-1}).
\end{aligned}$$

The second equality follows from the law of iterated expectations and the fourth equality follows from the definition of κ_{ik}^l.

The l period-ahead causal effect is a function of the average of the *information effect* and the *competition effect*, weighted by the frequency with which either would arise if k were kept in the network until period $l-1$, through the term $\kappa_{ik}^l P(\mathbf{s}_{l-1})$. Loosely speaking, if states where k would have been an information source, $s_{il-1} = 0, s_{kl-1} = 1$ are more likely and the competition and information effects are of comparable magnitudes, the information effect dominates and this term is negative. However, removing k from the network affects the employment probability of all members of the connected set that includes k, not just agent i.[2] This leads to an indirect *insurance effect* on i, captured by the term $\big[P_{-k}(\mathbf{s}_{l-1}) - P(\mathbf{s}_{l-1})\big]P_{-k}(s_{il} = 1|\mathbf{s}_{l-1})$. For example, suppose k is employed in period $l = 0$ and j is a common connection of i and k who is unemployed in $l = 0$. Agent j is less likely to be employed one period after losing the connection, $P_{-k}(s_{j1} = 1) < P(s_{j1} = 1)$. Intuitively, this affects i because agent j is now a worse source of information

[2] In period 1, the employment of agents within two links of k may be affected, in period 2, agents within three links of k may be affected, and so forth.

for i in period $l = 1$, should i find themselves unemployed. On the other hand, if both k and j are unemployed in $l = 0$, then j is more likely to be employed one period after losing the connection, $P_{-k}(\mathbf{s}_{j1} = 1) > P(\mathbf{s}_{j1} = 1)$. The insurance effect therefore captures the change in i's network's tendency to insure them against negative employment shocks in period $l-1$ when k is removed from the network.

The insurance effect is a function of $P(\mathbf{s}_{l-1})$. From the law of total probability and the Markov property, for $l \geq 2$ we can write:

$$P(\mathbf{s}_{l-1}) = \sum_{\mathbf{s}_0} \cdots \sum_{\mathbf{s}_{l-2}} P(\mathbf{s}_{l-1}|\mathbf{s}_{l-2}) \times \cdots \times P(\mathbf{s}_0),$$

with an analogous derivation for $P_{-k}(\mathbf{s}_{l-1})$, where $P_{-k}(\mathbf{s}_0) = P(\mathbf{s}_0)$. Each of these transition probabilities is the result of a combinatorial problem which in turn depends on the topology of k's entire connected set. The transition probabilities therefore do not admit a general, tractable closed-form expression. By extension, the difference $P_{-k}(\mathbf{s}_{l-1}) - P(\mathbf{s}_{l-1})$ also does not admit a general expression. However, one can prove by example that the sign of $\sum_{s_{l-1}} \big[P_{-k}(\mathbf{s}_{l-1}) - P(\mathbf{s}_{l-1})\big] P_{-k}(s_{il} = 1|\mathbf{s}_{l-1})$ is ambiguous. Therefore, the sign of the insurance effect is ambiguous and a function of the network topology and the initial distribution of state vector, $P(\mathbf{s}_0)$.

While it is not possible to make general statements about the sign of τ_{ik}^l without further assumptions on the structure of the network, the 1-period ahead effect is a special case. At this horizon, there is no insurance effect, since $P_{-k}(\mathbf{s}_0) = P(\mathbf{s}_0)$. The 1-period ahead causal effect is a function only of the information and competition effects, and can be written as follows:

$$\tau_{ik}^1 = \sum_{\mathbf{s}_0} \kappa_{ik}^1(\mathbf{s}_0) P(\mathbf{s}_0),$$

where

$$\kappa_{ik}^{1}(\mathbf{s}_0) =$$

$$\begin{cases} 0, & s_{i0} = 1 \\ -(1-b)(1-a)q_{k1}\prod_{\substack{j\neq k, j:g_{ij}=1 \\ s_{j0}=1}}(1-q_{j1}), & s_{i0}=0, s_{k0}=1 \\ (1-b)(1-a)\prod_{\substack{j\in I_{-k} \\ s_{j0}=1}}(1-q_{j1})\left[\prod_{\substack{j\in I_k \\ s_{j0}=1}}(1-q_{j1}) - \prod_{\substack{j\in I_k \\ s_{j0}=1}}(1-q_{j1}^{-k})\right], & s_{i0}=0, s_{k0}=0. \end{cases}$$

Finally, while the sign of the l period-ahead causal effect for any individual i is indeterminate, a reasonable conjecture, based on the results in Calvó-Armengol and Jackson (2004) and Calvó-Armengol (2004), is that, for a given network structure, the average steady-state value of $\lim_{l\to\infty}\tau_{ik}^{l}$ across individuals in k's connected set is negative. Intuitively, the information available locally to some agent i may increase when k is removed from the network, if the competition effect is sufficiently large or sufficiently common and the insurance effect is small or positive. However, provided the removed individual k would have been employed sufficiently often in a steady state, removing an individual from the network decreases the total information available to the network, suggesting that, in the aggregate, the members of the network will be worse off.

A.3 Simulations

To explore how the l period-ahead causal effect varies with the network topology around agents i and k, I conduct a set of simulations. I first generate a network with 100 nodes (agents), a single component, and an average degree of 3.9.[3]. The network is deliberately designed to contain two large clusters of nodes; professional networks as I construct them in Section 3 of the paper tend to exhibit high degrees of clustering, as all individuals sharing a workplace in a given year are linked. Clustering is also a general feature of professional networks (Newman and Park, 2003). The network is shown in Figure C.13.

To simulate the employment effect of losing a connection, I choose an agent k to remove from the network, and set the initial employment state of the removed individual k and a focal, treated, individual i with whom they are connected. I randomly draw the initial employment

[3]The average degree is lower than what is typically observed in real-world networks. The simulated effects will therefore be larger than what can be expected in a real-world network

state of all other individuals in the network $j \neq i, k$, setting $P(s_{j0} = 1) = \frac{a}{b+a}$, $a = 0.3$, and $b = 0.05$. Next, I simulate 1000 draws of job finding and job separation for each individual over six periods, given a and b. I focus on short- to medium-run outcomes, $l \leq 6$, rather than long-run steady-state outcomes, to align with the event studies in Section 4 of the paper. For each draw, I randomly order each agent's connections in each period; if an agent receives a job offer and does not use it, they pass in on to their highest-ranked unemployed connection in that period. I use the realisations of the random draws to simulate the evolution of the employment vector over the six-year horizon for the 1000 draws under the two alternative treatment arms: with k in the network and k removed from the network. I calculate the l period-ahead treatment effect τ_{ik}^{l} as the average in period l of the difference in i's employment status between the two treatment arms. I also calculate the share of draws where this difference is positive. I repeat this experiment for 100 draws of the initial employment state of all other individuals $j \neq i, k$.

I report the average effects in Figure C.14. As shown analytically, the effect is zero in $l = 1$ when $s_{i0} = 1$. When $s_{i0} = 0$, there is a weakly positive competition effect when $s_{k0} = 0$ and a negative information effect when $s_{k0} = 1$. However, the medium-term effect is negative over all configurations, with τ_{ik}^{l} ranging between zero and -4 percentage points. The ranking of medium-term effects is similar and stable over the choices of initial employment status, suggesting that six periods comes close to approximating the long-run effect. Agent i's degree has a first order importance: the medium-term simulated effects are ranked in order of i's degree; when i's degree is sufficiently high, the effect is statistically indistinguishable from zero. When i's degree is low, the medium-term effects are not ranked on either k's degree or the overlap between i and k's connections. Instead both k's degree and the overlap between i and k's connections matter. Both connections with a low degree and connections with a high degree and low overlap (*weak ties* in the sense of Granovetter, 1973)) have negative medium term effects of a similar magnitude.

Given the chosen parameter values, there are no network configurations under which the medium-run effect of losing a connection is positive. The share of simulations where the medium-run effect is positive is also small, less than one per cent. This is to be expected; when employment is generally high in a network, network members rarely compete for information and the information effect dominates the competition and insurance effects. I therefore re-run my simulations with $b = a = 0.3$. These parameter values do not correspond to any real-world

aggregate job finding and separation rates. However, they may correspond to annual job finding and separation rates for certain types of workers with low employment rates (Hall and Kudlyak, 2019; Doniger, 2026), particularly given that unemployment and non-participation are treated as a single state in the model. Medium-term positive effects are now slightly more common, particularly when k has a high degree and a high overlap with i's connections, as shown in Figure C.15. For this network configuration, the average effect is also positive though insignificant for all initial conditions.

B Online appendix: Data details

B.1 Variable definitions

Below, I define the variables used to estimate treatment propensity scores via post-double selection and heterogeneous treatment effects via the GRF. Appart from dyadic characteristics, all variables are defined at the level of an individual, who is categorised as either a focal individual or a past coworker.

B.1.1 Dyadic characteristics

An individual's set of coworkers is defined using matched worker-firm register RAMS provided by Statistics Sweden. For each dyad, I measure the *Number of coworkers* of either individual in the last year working together in the last common workplace. *Number of employed neighbours* is a count of the number of individuals who are past coworkers of both dyad members and who are employed in $t-1$. *Redundant connection* is an indicator that is equal to one if the the focal worker has multiple past coworkers employed at the workplace of the past coworker in $t-1$. *Years together* is a count of the years spent working together during the network formation phase, which has a maximum of five, *Years since coworking* is a count of the number of years since the focal individual and past coworker last worked together. *Focal individual moved* is an indicator for whether the focal individual has left the common workplace by year $t-1$; by construction, the past coworker has always left this workplace. I record the industry of the most recent common workplace at both the one- and two-digit level, using the SNI2002 codes. These codes are available for the period 2002–2010, for other years I rely on crosswalk provided by Statistics Sweden. Finally, I record the county (*län*) in which the most recent workplace was located.

B.1.2 Demographics

For both the focal individual and the past coworker, I draw data on basic demographic characteristics from the Louise/SYS register produced by Statistics Sweden. *Age* measures age in years in $t-1$, *Female* is an indicator for being female, and *Immigrant* is an indicator for being foreign-born. I define an *Education* variable with five categories, derived from detailed data on

school completion. I also define an indicator for *low education*, equal to one for individuals with less than a high school education.

B.1.3 Labour market history

Labour market history is likely both an important predictor of exposure to the unexpected death of a former coworker and an important determinant of future outcomes. It will also influence the size of an individual's network in $t-1$. For both focal individuals and past coworkers, use the RAMS worker-firm register to measure the *Number of establishments* worked at during the network building phase, *Tenure* in one's establishment in $t-1$ (ranging from one to six, equal to zero for unemployed individuals), as well as labour market *Earnings* (wages plus business income) in 2018 SEK for years $t-6$ to $t-1$ and an *Employment* indicator for each of those years, equal to one if earnings are positive.

In addition to the detailed measures of labour market history described above, I also income more summary measures: an indicator for being a *High earner* in $t-1$ (above-median earnings), an indicator for having experienced any *Past non-employment* during the network-building phase, and an individual's *Average employment* during the network building phase. This is to ensure that variable selection algorithms such as the LASSO or the GRF, which select based on the strength of the association between a single variable and an outcome, do not miss the relationship between, say, past employment and future earnings, because of noise introduced by measuring employment separately in each year.

B.1.4 Industry affiliation

I include distinct measures of affiliation with different industries for the focal individual and the past coworker, again using the SNI2002 codes recorded in the linked employer-employee register RAMS. For the focal individual, I measure the number of *Years worked* in each one- and two-digit industry during the network building phase, to capture their industry-specific human capital. For the past coworker, I include a set of indicators for being employed in each one- and two-digit industry in year $t-1$ (where unemployment is treated as a separate industry). This captures potential differences in the importance of referrals and the value of connections across different industries.

B.1.5 Network characteristics

Network theory suggests that one of the main determinants of the value of a connection, measured as the probability that information will flow within a dyad, is be the local topology of the network around that dyad (Calvó-Armengol, 2004) and the economic outcomes of neighbouring individuals, (Calvó-Armengol and Jackson, 2004). For both the focal individual and their past coworker, I measure the *Number of employed contacts*, the number of past coworkers who are working at a new workplace in $t-1$, and the *Number of non-employed contacts*, the number of past coworkers who are in non-employment in $t-1$. I also measure the *Number of connected workplaces*, then number of distinct new workplaces where their past coworkers are employed in $t-1$ and the *Number of connected firms*, the number of distinct new firms where their past coworkers are employed in $t-1$.

B.1.6 Prior exposure to coworker deaths

I do not place any restrictions on mortality among either the focal individual or past coworker's set of past coworkers, as defined in $t-1$. For both the focal individual and the past coworker, I therefore also define an indicator for *At least one prior coworker death*, equal to one if at least one past coworker, defined in $t-1$ dies of any cause before year t, and an indicator for *Multiple prior* coworker deaths, equal to one if more than one past coworker dies fo any cause before year t.

B.2 Propensity score matching

B.2.1 Propensity score estimation via post-double selection

Here I describe the details of the post-double selection procedure for estimating treatment propensity. I repeat the procedure for each year. This accounts for possible differences in the underlying distribution of causes of death over years as well as business cycle fluctuations that may affect the value of a connection.

Predicting treatment: To select predictors of treatment, I fit a logistic regression for treatment status to the treated dyads and a 1 per cent sample of control dyads. I select the number of regressors to maximise the AUC criterion, via five-fold cross validation. I focus on AUC as the

treatment is rare. In such cases, MSE can be a poor measure of model fit and minimising MSE can lead to degenerate solutions where all observations are simply classified as non-treatment. Maximising the AUC improves performance, limiting the rate of both false positives and false negatives.

Predicting outcomes: To select predictors of outcomes, I fit a linear regression for the change in earnings between $t-1$ and $t+2$ to the treated individuals and a new 1 per cent sample of potential control individuals. I select the number of regressors to minimise the prediction MSE, via five-fold cross validation. This LASSO estimation only includes covariates measured at the level of the focal individual.

Estimating the propensity score: Next, I draw a new 1 per cent sample of control dyads. Using the sampled dyads and the treated dyads, I fit a logistic regression for treatment status on the union of the variables selected by the two LASSOs. Taking the union across years, the post-double selection procedure selects 494 unique variables. I report all variables selected in at least 10 years in Table C.2. Only three predictors of treatment are selected in all years: the focal worker's age and average past employment and the coworker's past employment. In general, the post-double selection algorithm selects many characteristics of the focal individual that are predictors of the outcome only, reflecting the greater variation in outcomes than in treatment status. However, including these predictors turns out to have little effect on the estimated propensity score. To show this, I also estimate the propensity score using only the selected predictors of the treatment status, on the same random sample. For the treated dyads and matched controls, the correlation between this post-LASSO propensity score and the post-double selection propensity score used in the matching below is 0.99.

B.2.2 Matching on the estimated propensity score

For all treated dyads and all control dyads not used either to select predictors or to fit the post-double selection propensity score model, I predict the treatment probability using my estimated propensity score model. Using this restricted set of observations ensures that no data from the potential matched control dyads was used to train the propensity score model, reducing the risk of overfitting. I match without replacement within strata within years. The matching strata are defined using three variables: focal individual's employment status in $t-1$, past coworker's

employment status in $t-1$, and an indicator equal to one if the past coworker is a redundant connection, in the sense that another of the focal individual's past coworkers is employed at the same workplace as the past coworker (equal to zero if the past coworker is unemployed). For each treated dyad, I select 10 control dyads, to increase power both to identify average effects and to explore heterogeneity in treatment effects.

C Online appendix: Additional results

Table C.1: Summary statistics for treated and control dyads

	Controls (unmatched) (1)	Controls (matched) (2)	Treated (matched) (3)
N dyads	650,671,842	975,520	97,552
N focal individuals	36,760,844	819,657	97,552
N coworkers	27,338,695	878,376	4,129
Dyad characteristics			
N coworkers, shared workplace	49.9	47.7	49.0
N employed neighbours	22.9	21.3	22.2
Redundant connection	0.160	0.125	0.125
Years together in [t-6,t-2]	1.53	1.51	1.49
Years since coworking	3.06	3.05	3.07
Focal individual moved	0.642	0.624	0.630
Focal individual			
Demographics			
Age	36.2	36.4	36.3
Female	0.495	0.395	0.385
Immigrant	0.104	0.108	0.106
Education: compulsory or less	0.148	0.184	0.187
Education: high school $\leq$2 years	0.291	0.339	0.340
Education: high school 3 years	0.254	0.250	0.246
Education: tertiary $\leq$2 years	0.156	0.125	0.124
Education: tertiary $\geq$3 years	0.152	0.103	0.103
Labour market history			
N establishments in [t-6,t-2]	2.03	1.98	2.00
Tenure (years)	2.72	2.76	2.75
Earnings, t-1 (SEK '000)	195.3	188.7	189.0
Earnings, t-2	188.4	183.3	183.6
Earnings, t-3	179.8	175.6	176.1
Earnings, t-4	170.0	166.6	167.2
Earnings, t-5	159.9	156.0	156.4
Earnings, t-6	149.3	145.5	145.9

continued on next page

Table C.1 – continued

	Controls (unmatched) (1)	Controls (matched) (2)	Treated (matched) (3)
Employed, t-1	0.876	0.866	0.866
Employed, t-2	0.888	0.880	0.879
Employed, t-3	0.892	0.885	0.885
Employed, t-4	0.884	0.877	0.876
Employed, t-5	0.861	0.853	0.851
Employed, t-6	0.830	0.820	0.820
Network characteristics			
N employed contacts	60.1	54.6	55.9
N non-employed contacts	21.8	21.6	22.0
N connected workplaces	46.0	41.9	43.1
N connected firms	42.9	39.6	40.8
Prior exposure to coworker death			
At least one prior coworker death	0.328	0.356	0.351
Multiple prior coworker deaths	0.096	0.112	0.117
Past coworker			
Demographics			
Age	35.1	35.8	36.5
Female	0.491	0.231	0.212
Immigrant	0.108	0.121	0.111
Education: compulsory or less	0.140	0.201	0.250
Education: high school $\leq$2 years	0.265	0.320	0.325
Education: high school 3 years	0.269	0.251	0.256
Education: tertiary $\leq$2 years	0.164	0.124	0.089
Education: tertiary $\geq$3 years	0.161	0.104	0.080
Labour market history			
N establishments in [t-6,t-2]	2.39	2.16	2.24
Tenure (years)	1.45	1.11	1.06
Earnings, t-1 (SEK '000)	165.2	128.1	117.8
Earnings, t-2	161.1	129.6	121.1
Earnings, t-3	153.7	130.3	123.6
Earnings, t-4	145.4	128.9	124.1

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Table C.1 – continued

	Controls (unmatched) (1)	Controls (matched) (2)	Treated (matched) (3)
Earnings, t-5	137.1	126.1	121.1
Earnings, t-6	128.7	121.8	118.8
Employed, t-1	0.779	0.613	0.613
Employed, t-2	0.835	0.690	0.698
Employed, t-3	0.846	0.739	0.755
Employed, t-4	0.841	0.760	0.771
Employed, t-5	0.821	0.760	0.759
Employed, t-6	0.791	0.740	0.752
Network characteristics			
N employed contacts	75.6	64.8	65.3
N non-employed contacts	27.5	25.5	25.7
N connected workplaces	55.0	46.7	47.6
N connected firms	50.5	43.5	44.4
Prior exposure to coworker death			
At least one prior coworker death	0.329	0.352	0.350
Multiple prior coworker deaths	0.098	0.109	0.113

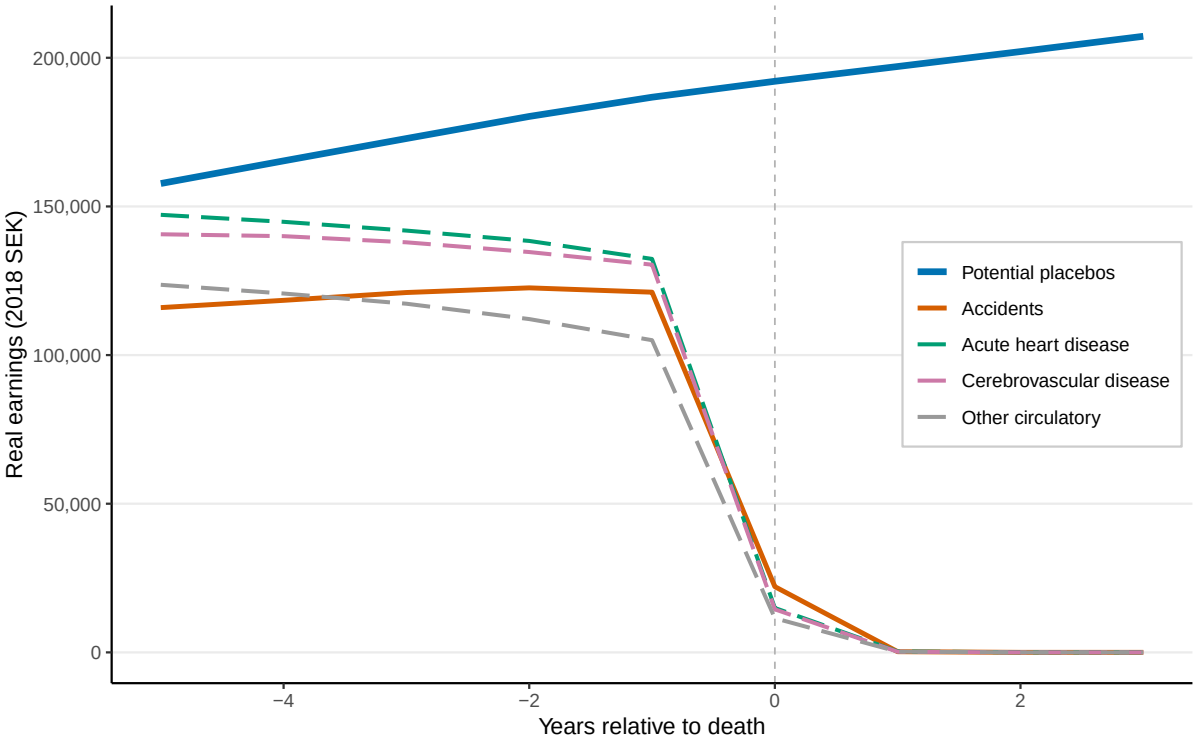
Note: Mean values of all covariates for unmatched control dyads (Column 1), matched control dyads (Column 2), and treated dyads (Column 3). Earnings are deflated and expressed in thousands of 2018 SEK. See the data appendix for variable definitions.

Table C.2: Predictors selected in at least 10 years

	N years selected	
	Treatment or outcome	Treatment only
Age	20	20
Coworker: mean employment, t-6 to t-2	20	20
Mean employment, t-6 to t-2	20	20
Past non-employment	20	18
Earnings, t-1	20	0
Earnings, t-1 × Female	20	0
Education: tertiary, 2 years or less	20	0
Coworker: female	19	19
Education: tertiary, 3+ years	19	0
Earnings, t-1 × Employed, t-1	18	0
Earnings, t-1 × Past non-employment	17	0
Earnings, t-6	17	0
Earnings, t-5	15	0
Earnings, t-1 × Employed, t-2	14	0
Earnings, t-3	14	0
Coworker: employed, t-2	13	13
Employed, t-2	13	8
Earnings, t-1 × Low education	13	0
Earnings, t-4	12	0
Education: tertiary, 2 years or less × Past non-employment	12	0
N coworkers, shared workplace	11	11
Coworker: employed, t-1	11	11
N establishments, t-6 to t-2	11	3
Education: tertiary, 3+ years × Past non-employment	11	0
Employed, t-1	10	10
Industry: computer/IT (SNI2002 72)	10	0

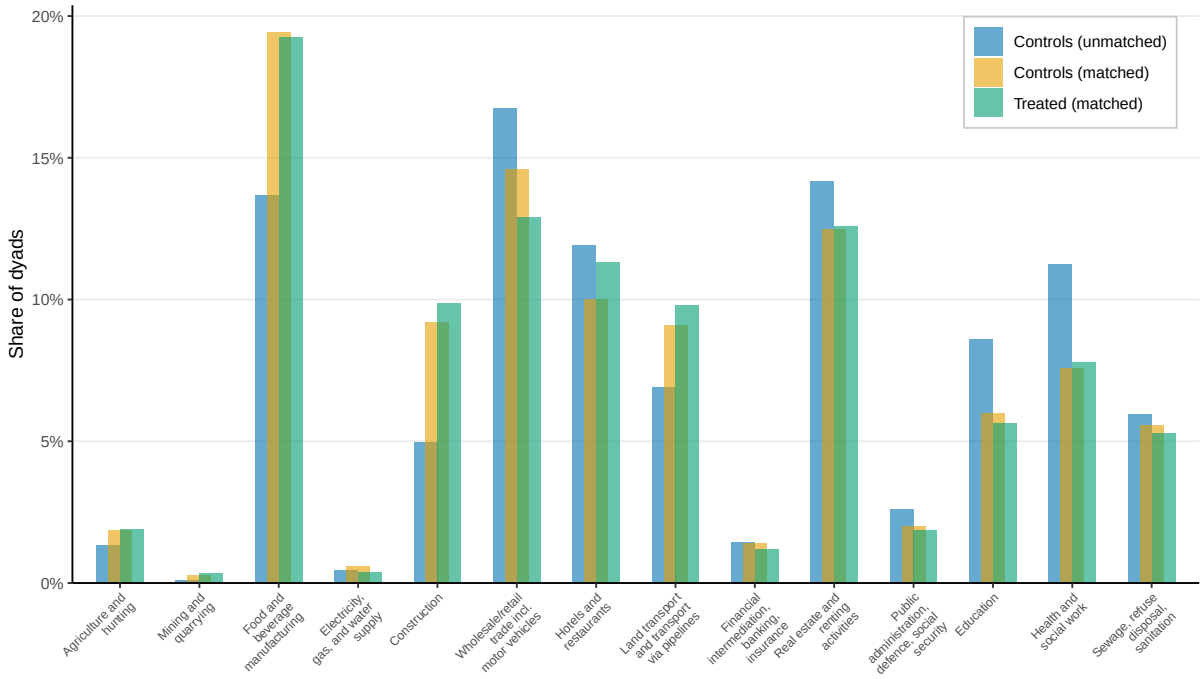
Notes: Variables selected as predictors of either treatment or change in earnings in at least 10 years. Variables refer to focal individuals unless otherwise stated.

Figure C.1: Earnings trajectories; deceased and comparison individuals



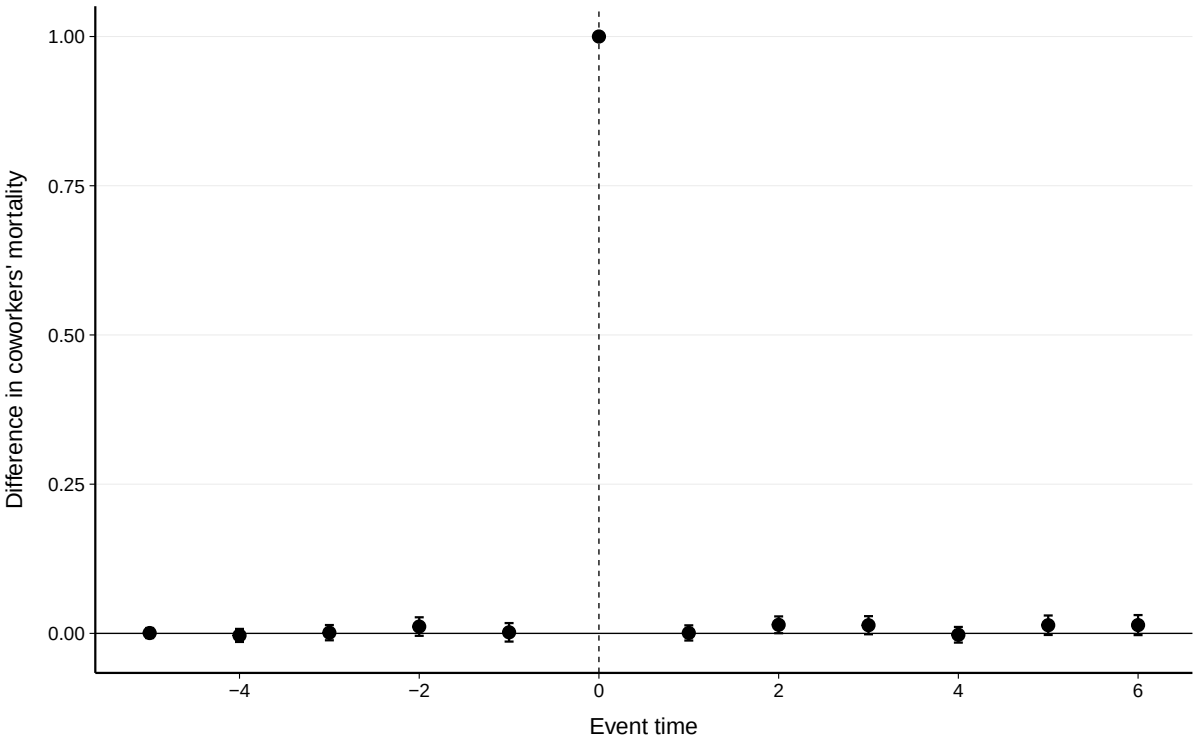
Notes: Annual earnings for individuals satisfying sample restrictions who die in 1991-2010. Comparison earnings trends are calculated for individuals who not die in each observation year, years are then pooled. Potential placebos are connections of control individuals.

Figure C.2: Shared past industry



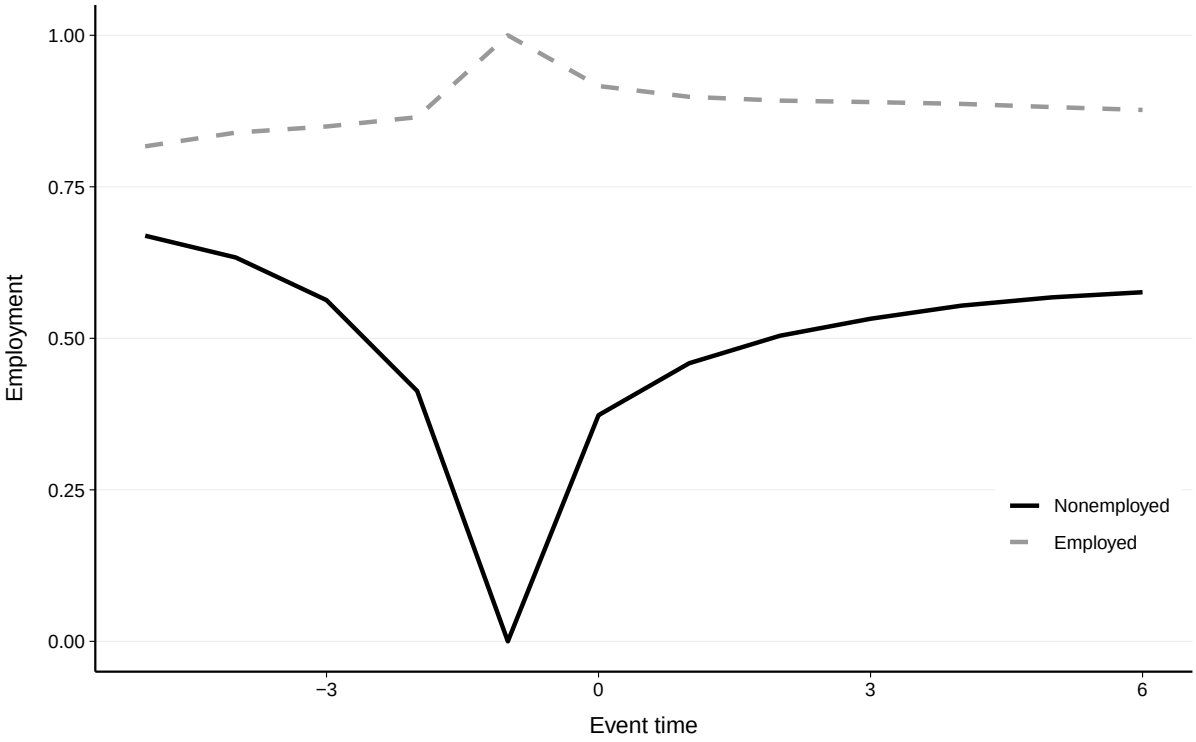
Notes: Distribution over one-digit industries of the workplace where the focal individual last worked with their past coworker, SNI2002 classification. Extra-territorial organisations and Private households have been omitted due to too few cases.

Figure C.3: Relationship between treatment and total past coworker mortality



Notes: Simple differences between treated and control individuals in total deaths (all causes) among past coworkers. Past coworkers are defined in $l = -1$ based on workplaces in years $l \in [-6, -2]$ and are not assumed to be alive in $l = -1$. Standard errors are clustered by past coworker by event year, 95 per cent confidence intervals are shown.

Figure C.4: Past coworker's employment status

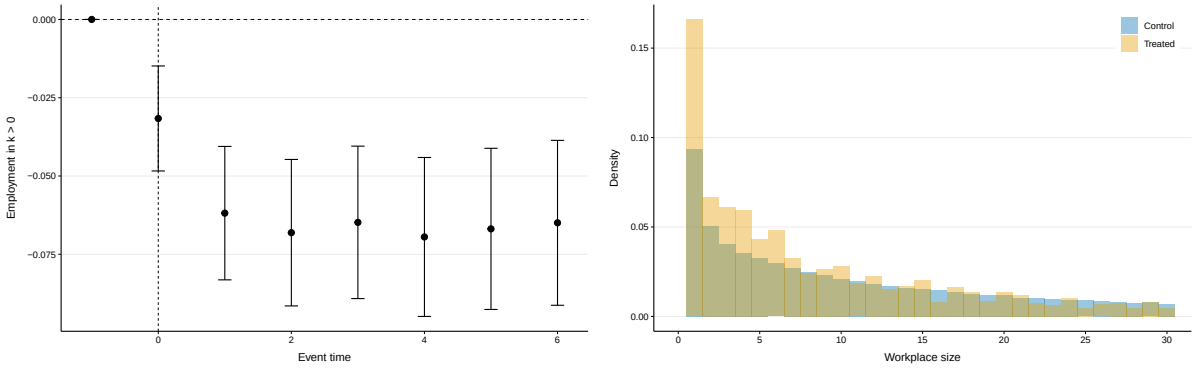


Notes: Employment rate of past coworkers in event year l, by past coworker's employment status in $l = -1$. the sample is restricted to placebo past coworkers; deceased past coworkers' employment rate is zero from $l = 1$ by construction.

Figure C.5: Unexpected death of an employee and workplace survival

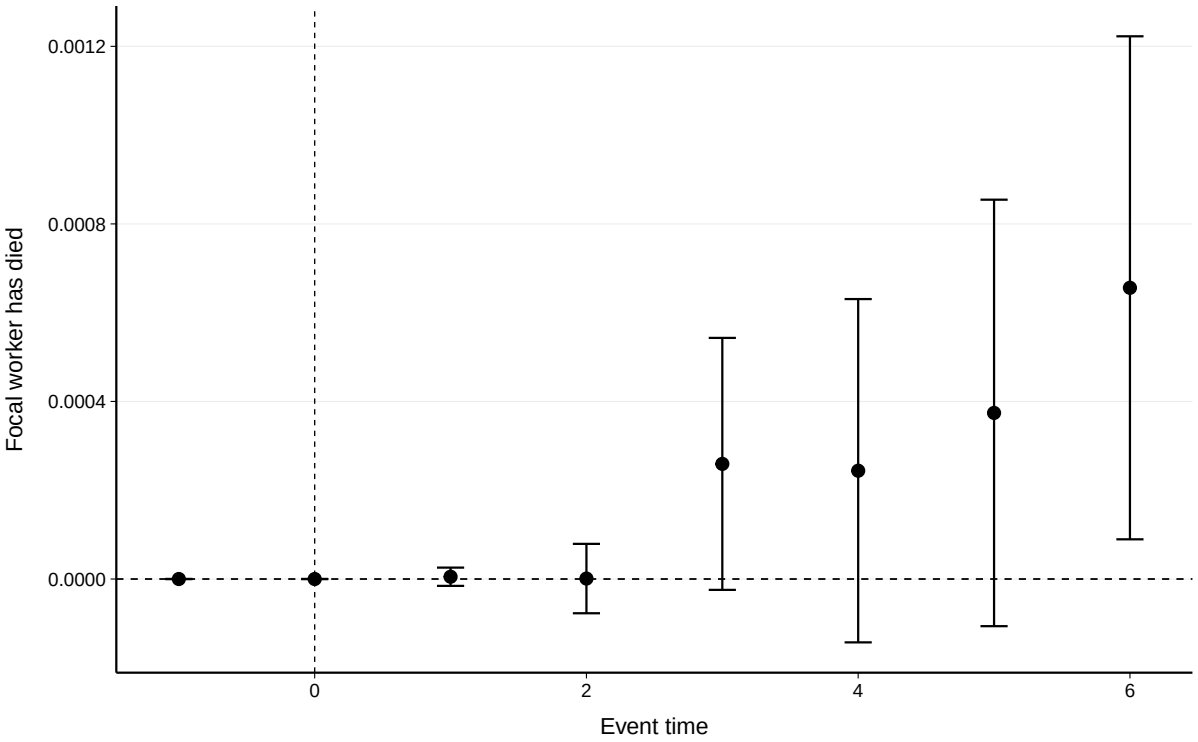
(a) Workplace exists

(b) Workplace size, $t - 1$



Note: Panel (a) reports event study estimates where the outcome is an indicator for the workplace of the former coworker in $t - 1$ employing at least one worker in event year l. The sample is restricted to dyads where the past coworker was employed in year $t - 1$, standard errors are clustered by past coworker and event year, 95 per cent confidence intervals are reported. Panel (b) reports that truncated distribution of past coworkers' workplace sizes in $t - 1$ for workplaces that no longer employ anyone in $t + 3$. The distribution is truncated at 30. 22 per cent of treated dyads and 38 per cent of control dyads where the past coworker was employed in $t - 1$ and the workplace was closed in $t + 3$ are above the threshold.

Figure C.6: Focal worker's mortality

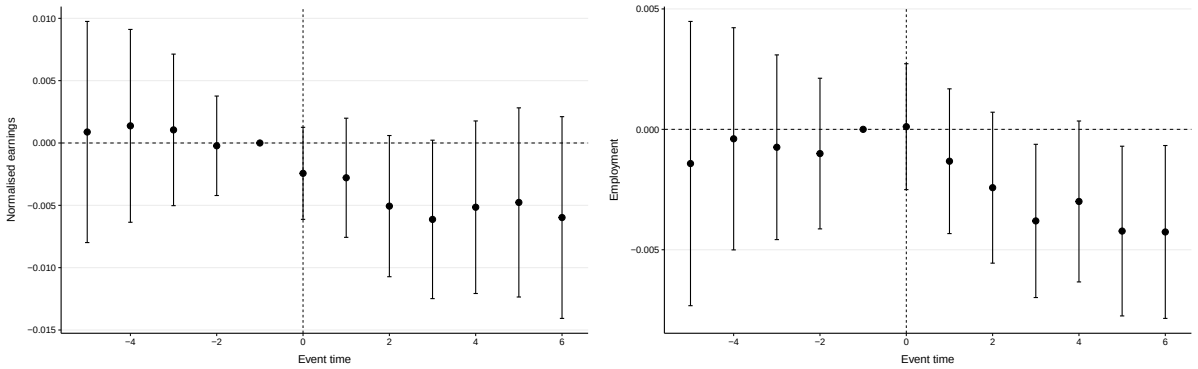


Notes: Estimate of the effect of exposure to a past coworker death on own mortality. The outcome is an indicator for having died by the end of year l. The sample is obtained by matching treated dyads to control dyads, as described in the text. Standard errors are clustered by past coworker by event year, 95 per cent confidence intervals are reported.

Figure C.7: Labour market effect of unexpected past coworker deaths

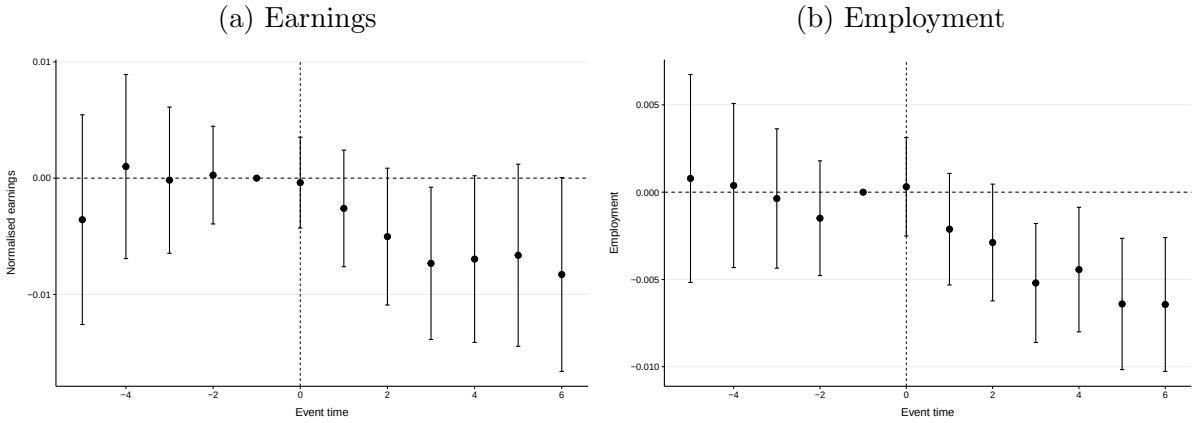
(a) Earnings

(b) Employment



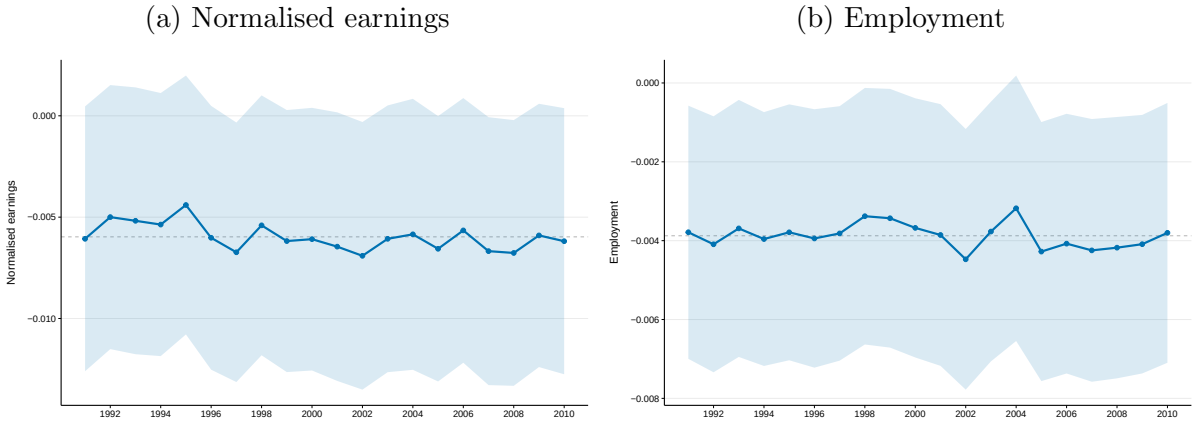
Note: The figures report event-study estimates of the effect of unexpected deaths of past coworkers, occurring during 1991–2010. The sample is restricted to include only treated and control individuals alive in $t+6$. Real annual labour earnings, measured in 2018 Swedish crowns, are normalised by average earnings in the control group in each event year by death year (Panel A). Employment is a dummy for having positive labour earnings during the calendar year (Panel B). Coefficients are estimated on the matched sample, using Equation (2). Standard errors are clustered by past coworker by event year, 95 per cent confidence intervals are reported.

Figure C.8: Labour market effect of unexpected past coworker deaths, alternative matching



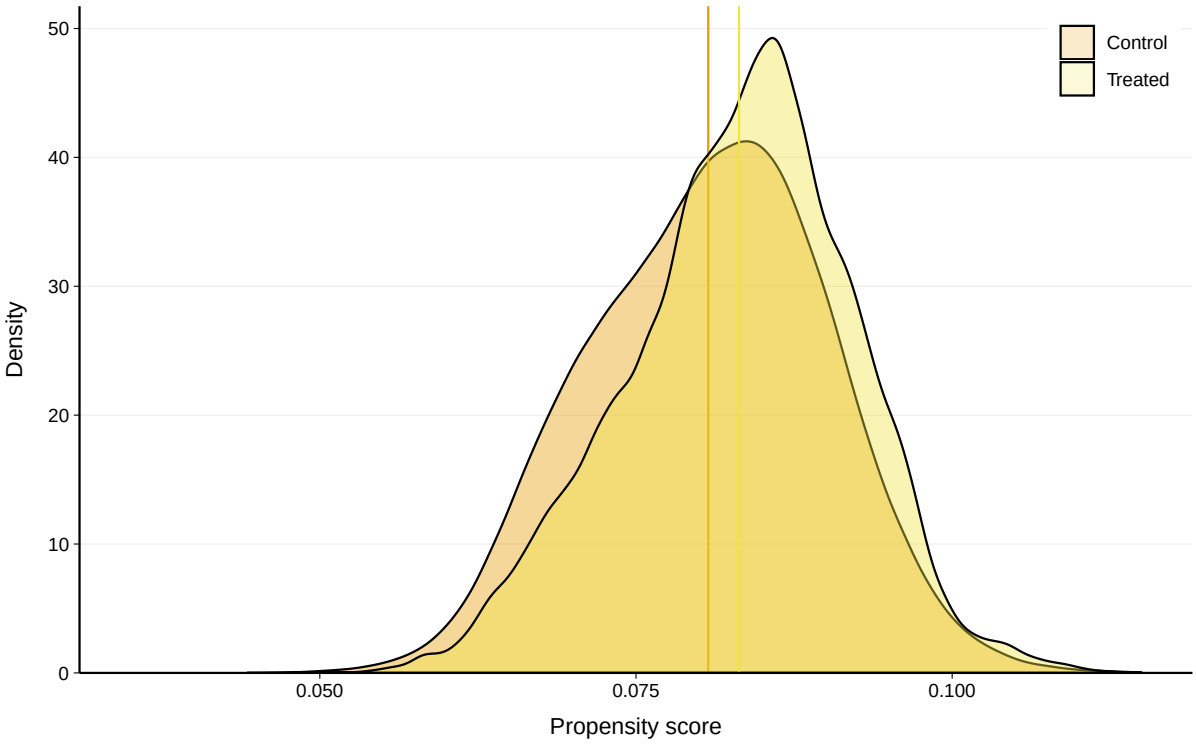
Note: The figures report event-study estimates the effect of unexpected deaths of past coworkers, occurring during 1991–2010. Real annual labour earnings, measured in 2018 Swedish crowns, are normalised by average earnings in the control group in each event year by death year (Panel A). Employment is a dummy for having positive labour earnings during the calendar year (Panel B). Coefficients are estimated on the alternative matched sample, using Equation (2). Standard errors are clustered by past coworker by event year, 95 per cent confidence intervals are reported.

Figure C.9: Robustness to omitting individual years



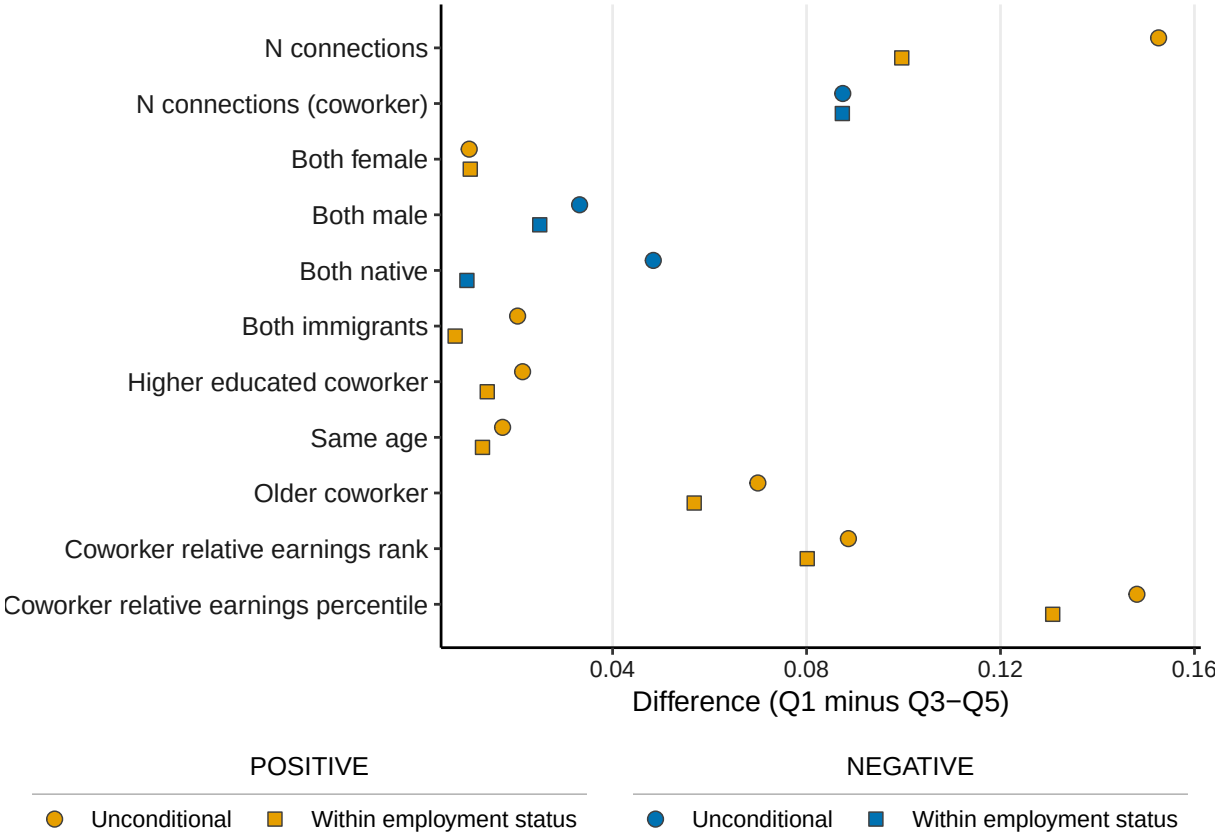
Note: The figures report difference-in-differences estimates of the effect of unexpected deaths of past coworkers on outcomes measured the year before the event and three years after the event. Each reported coefficient corresponds to a different subsample of the original matched sample, obtained by dropping treated dyads and matched controls in each calendar year in turn. The average effect including all years is shown as a dashed line. Real annual labour earnings, measured in 2018 Swedish crowns, are normalised by average earnings in the control group in each event year (Panel A). Employment is a dummy for having positive labour earnings during the calendar year (Panel B). Standard errors are clustered by past coworker by event year, 95 per cent confidence intervals are reported.

Figure C.10: Forest-estimated treatment propensities



Notes: Probability density of the estimated treatment propensities for the matched sample. The sample is split into five folds. For each held-out fold I fit a classification forest on the remaining folds to predict the treatment propensities in the held-out fold. Random sampling into folds is clustered by past coworker by event year. I plot the mean predicted treatment propensity for each group.

Figure C.11: Differences in constructed covariates

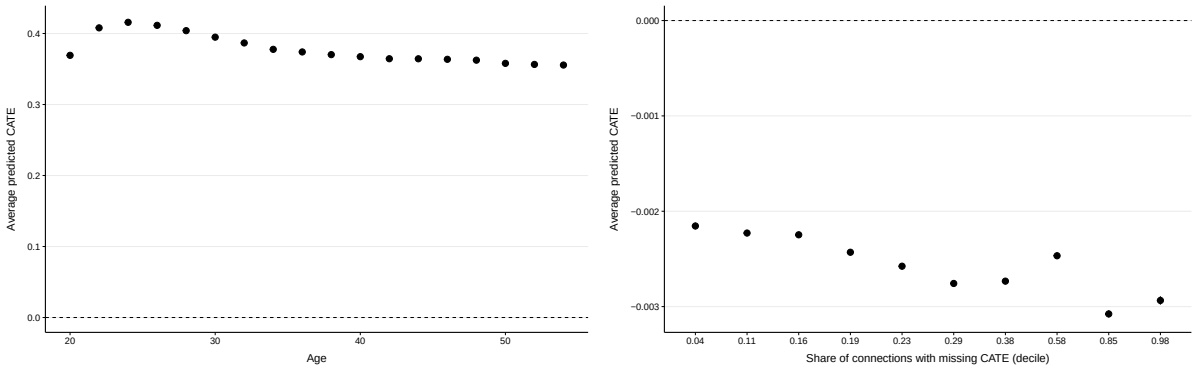


Notes: Absolute values of differences in average covariates between the bottom quintile of the predicted CATE distribution and quintiles 3–5. The sign of the difference is shown in the legend. The number of connections has been standardised to have mean 0 and standard deviation 1. Other variables are either binary measures of similarity or are measured on $[-1, 1]$ (relative rank/percentile). Education is measured using five categories, same age is an indicator for being within five years in age. Quintiles are defined by ranking CATEs either within fold, or within fold and employment status in $t-1$ for both focal individual and past coworker.

Figure C.12: Missing CATEs

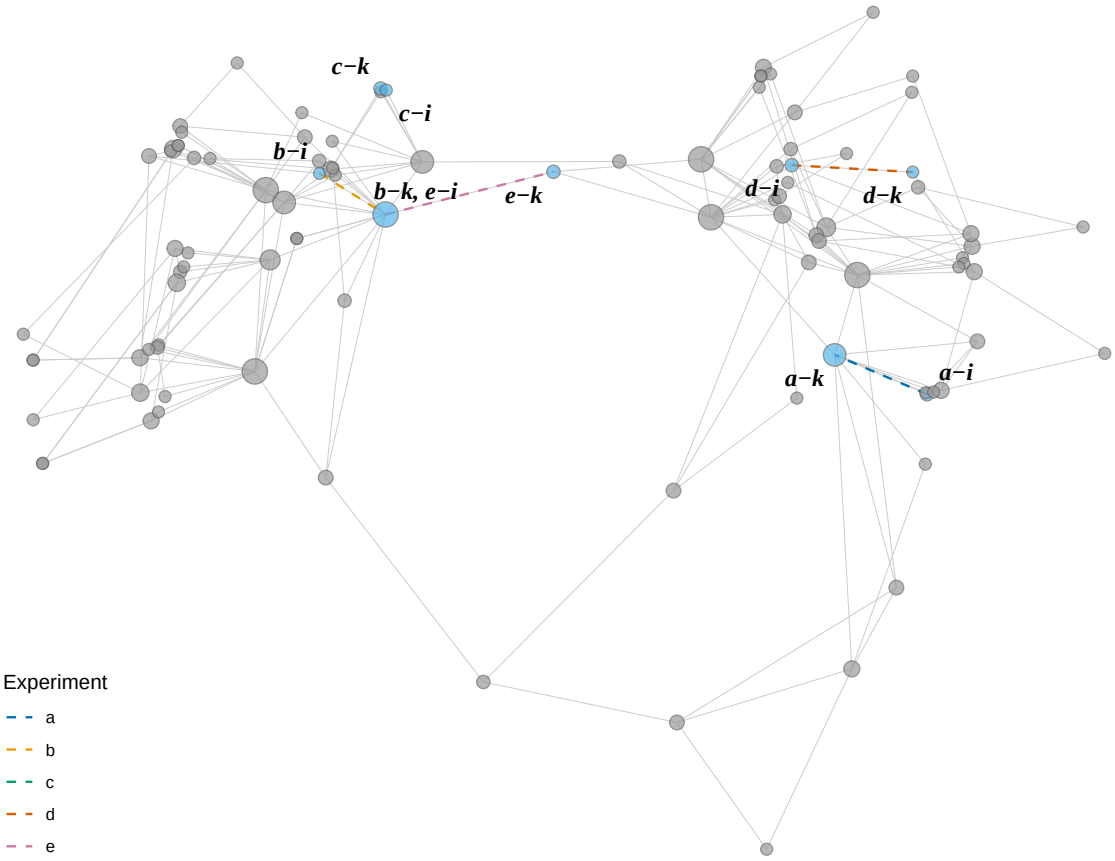
(a) By age

(b) Effect on average CATE



Notes: Share of connections missing a predicted CATE due to missing data. Age is grouped in two-year bins, I calculate and report the average share of missing connections per decile.

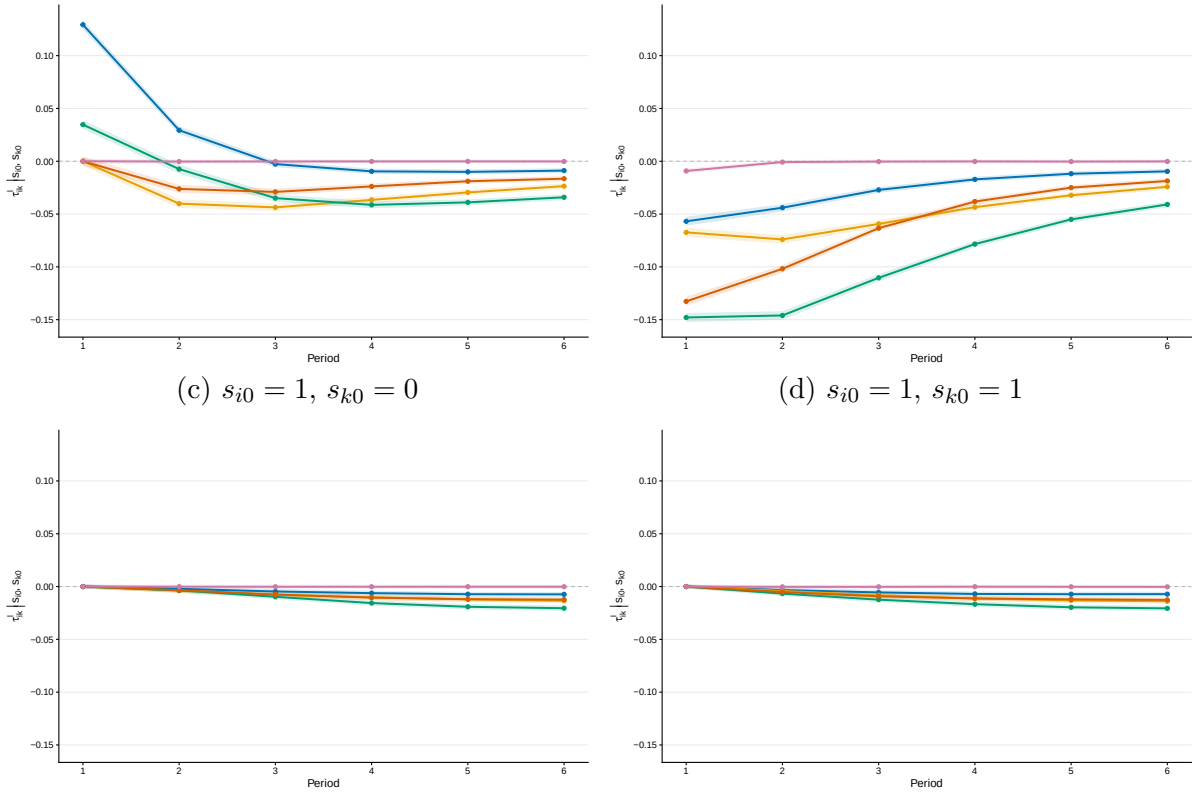
Figure C.13: Example of a professional network



Notes: A randomly generated social network, containing 100 nodes and two connected clusters, with average degree 3.9. Each experiment consists in removing an individual k from the network (i.e. removing all links to k) and studying the effect on a remaining individual i who was previously connected to k.

Figure C.14: Simulated l period-ahead employment effect, τ_{ik}^{l}

(a) $s_{i0} = 0,\ s_{k0} = 0$ (b) $s_{i0} = 0,\ s_{k0} = 1$



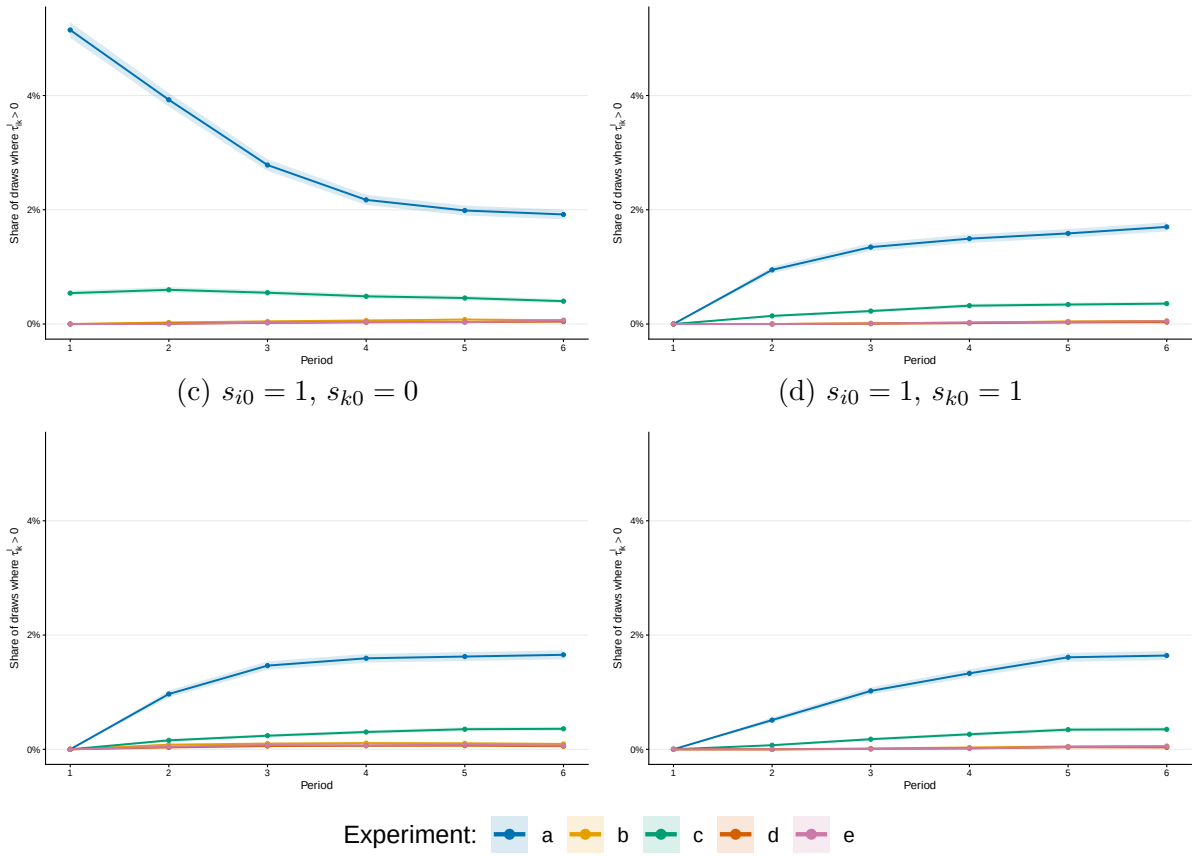
(c) $s_{i0} = 1,\ s_{k0} = 0$ (d) $s_{i0} = 1,\ s_{k0} = 1$

Note: Average simulated effect of losing a connection with $a = 0.3$, $b = 0.05$. The experiments are defined as follows, where $d(\cdot)$ denotes an agent's degree and $n(i,k)$ denotes the number of common neighbours of agents i and k. *experiment a*: $d(i) = 3$, $d(k) = 10$, $n(i,k)/d(i) = 0.67$; *experiment b*: $d(i) = 2$, $d(k) = 15$, $n(i,k)/d(i) = 0$; *experiment c*: $d(i) = 2$, $d(k) = 3$, $n(i,k)/d(i) = 0.5$; *experiment d*: $d(i) = 3$, $d(k) = 2$, $n(i,k)/d(i) = 0$; *experiment e*: $d(i) = 15$, $d(k) = 3$, $n(i,k)/d(i) = 0$. 95 per cent confidence intervals reported.

Figure C.15: Positive simulated l period-ahead employment effects, $\tau_{ik}^{l} > 0$

(a) $s_{i0} = 0$, $s_{k0} = 0$ (b) $s_{i0} = 0$, $s_{k0} = 1$

(c) $s_{i0} = 1$, $s_{k0} = 0$ (d) $s_{i0} = 1$, $s_{k0} = 1$



Note: Share of positive simulated effect of losing a connection with $a = 0.3$, $b = 0.3$. The experiments are defined as follows, where $d(\cdot)$ denotes an agent's degree and $n(i,k)$ denotes the number of common neighbours of agents i and k. *experiment a*: $d(i) = 3$, $d(k) = 10$, $n(i,k)/d(i) = 0.67$; *experiment b*: $d(i) = 2$, $d(k) = 15$, $n(i,k)/d(i) = 0$; *experiment c*: $d(i) = 2$, $d(k) = 3$, $n(i,k)/d(i) = 0.5$; *experiment d*: $d(i) = 3$, $d(k) = 2$, $n(i,k)/d(i) = 0$; *experiment e*: $d(i) = 15$, $d(k) = 3$, $n(i,k)/d(i) = 0$. 95 per cent confidence intervals reported.